Security Governance - Notes

IMPORTANT WARNING

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CHAPTER 1: Introduction

The Enterprise as a Complex System

Every organization exists to fulfil a **mission**, that is, the fundamental business objective that justifies its existence and defines what it delivers to the world. An energy provider exists to produce and sell energy to its customers; a manufacturer exists to produce and distribute goods; a bank exists to safeguard and move money. This mission is the anchor from which every other organizational choice descends, because once the objective is fixed, the enterprise must structure itself in a way that allows that objective to be reached reliably and repeatably.

From the outside, an enterprise can be understood as a kind of black box: it consumes inputs and produces goods or services that are offered to customers. Inside that black box, however, sits a highly structured machine composed of three interlocking ingredients. The first ingredient is **people** — the employees whose work actually moves the organization toward its objective. The second is **processes** — the structured sequences of activity through which work is carried out, formalized under the discipline of Business Process Management (BPM), in which people and technology transform inputs into outputs. The third is the **digital infrastructure** — the technological substrate of servers, networks, applications and data on which the people rely to perform their processes. The way these three are arranged depends heavily on scale: a small company tends to have a flat structure with few intermediaries, whereas a large enterprise develops a deep hierarchy with many layers of coordination.

This composition is precisely what makes the enterprise a **complex system**, and it is also the source of its first and most persistent vulnerability. Because the digital infrastructure is operated by human beings, the point of contact between people and technology becomes the weakest link in the whole arrangement. Managing the security of even a small company is therefore never a trivial task, and the difficulty grows sharply with organizational size and complexity.

The Concept of Governance

Among the many processes an enterprise runs, this treatment concentrates on **governance**. Governance, in its broadest sense, is the totality of the processes of governing — whether these are undertaken by a government, a market, or a network, and whether they act over a family, a tribe, a formal or informal organization, a territory, or across territories. Governance operates through the laws, norms, power, and language of an organized society. Expressed in operational terms, governance is the set of processes of interaction and decision-making among the actors involved in a collective problem, and these processes lead to the creation, reinforcement, or reproduction of social norms and institutions.

Two words distill the essence of this definition and should be held firmly in mind: **processes** and **norms**. To govern is to look at the business objective and to put in place everything necessary to reach it, which means identifying the processes required and then operating those processes so that the objective is achieved *in compliance with the surrounding environment.* Doing governance therefore always carries a double burden. It is not enough to design processes from a purely functional point of view, asking only whether they work; one must simultaneously keep the **compliance** dimension in view, ensuring that the processes respect the applicable norms. Those norms may be external laws, internal company rules, contractual obligations, or regulatory prescriptions. Governance without compliance is incomplete governance.

Enterprise, Corporate, and Business Governance

The most general form of this discipline is **enterprise governance**, which concerns how to govern the company as a whole. Formally, enterprise governance is the structure and the set of relationships that control, direct, or regulate the performance of an enterprise together with its projects, portfolios, infrastructure, and processes. In other words, it is the exercise of identifying the processes necessary to control the company itself. Enterprise governance is conventionally split into two complementary halves.

The first half is **corporate governance**, understood as the system of rules, practices, and processes by which a firm is directed and controlled. Corporate governance is essentially concerned with **compliance processes**: it is a general, mission-independent approach that guarantees the firm is being run in a controlled and accountable way, and it applies to any company regardless of what that company actually produces.

The second half is **business governance**, understood as the set of policies and business processes that determine the way the organization's business is actually run. Business governance is concerned with **performance processes**, and unlike corporate governance it is highly specific to the mission. Where corporate governance asks whether the firm is being directed and controlled properly in a generic sense, business governance asks whether the firm is performing well against its particular objective. The two are therefore distinguished cleanly: corporate governance is general and compliance-oriented, business governance is specific and performance-oriented.

The Hierarchical Structure and the Direct/Control Loop

An enterprise is typically organized as a **pyramid**. At the apex sit a small number of people who make broad, high-consequence decisions; those decisions then propagate downward through the structure, becoming progressively more concrete at each descent. This vertical arrangement gives rise to three well-known management levels, and the mechanics of corporate governance can be read directly off them.

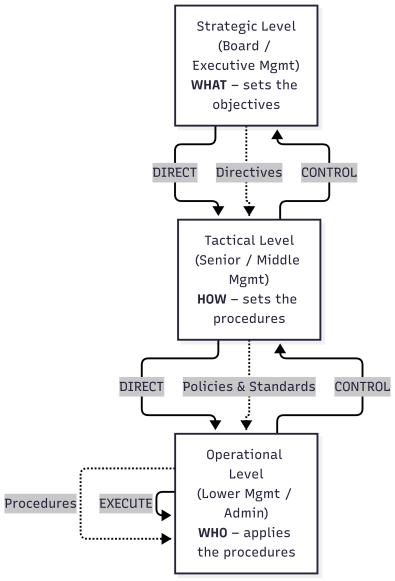
At the **strategic level** — populated by the Board of Directors and executive management — decisions about cybersecurity are taken at their most abstract. Directors formalize these decisions into **Directives**, documents that state the strategic objectives of the company. A directive at this altitude might declare that the infrastructure must be protected from external attackers, but it deliberately refrains from specifying *how* this is to be accomplished. The strategic level owns the **WHAT**: it sets the objectives.

At the **tactical level** — populated by senior and middle management — the mandate received from above is translated into more concrete objectives. Here the discussion turns to *how things should be done*, and the abstract goal begins to acquire shape. It is at this layer that one starts to identify the procedures needed to protect the assets, defining the **policies** and the **company standards** that will

govern behaviour. These standards are not necessarily imposed from outside by bodies such as ISO; they are frequently the internal standards the company chooses to adopt for itself. The tactical level remains a *meta* layer, in the sense that it defines the way things must be done without yet touching the machines. The tactical level owns the **HOW**: it sets the procedures.

At the **operational level** — populated by lower management and administration — the middle-level objectives are finally converted into practical reality. Here the abstract instruction to deploy, say, an identity management system becomes a concrete decision about who does exactly what, on which machine, with which configuration. This is the layer in which people put their hands on the systems. The operational level owns the **WHO**: it applies the procedures, and it is where things are actually done.

These three levels are bound together by a continuous loop composed of three motions. **Direct** is the downward flow: it translates big, abstract objectives into concrete implementation procedures as they descend from strategic to tactical to operational. **Execute** is the practical act of doing the work at the operational level. **Control** is the upward flow: it observes whether what is being done actually works as expected, and it feeds the resulting information back up to management. When the feedback arrives at the top, management obtains a snapshot of the real situation — an understanding of how the company is genuinely behaving — and on the basis of that snapshot it can decide to hold course or to change direction, re-drafting the objectives as needed. The following diagram captures this continuous circulation.



The single most important idea to extract from this picture is that **talking about governance means talking about the implementation of a continuous**

loop. Governance is not a one-off act of decision but a perpetual cycle of directing, executing, controlling, and re-directing.

Vulnerabilities as the Enemies of Governance

Once this loop is running inside a real company, it must contend with **vulnerabilities**, which are the natural enemies of the governance process. Vulnerabilities arise for two broad reasons. The first is that the technical infrastructure itself changes over time and inevitably contains bugs, errors, and even backdoors. The second, and more insidious, is the interaction between the processes and the infrastructure — and, above all, the interaction between *people* and the infrastructure. It bears repeating as a central lesson that the most dangerous moment in the whole governance enterprise is the moment at which people come into contact with the systems.

Risk as the Governing Metric

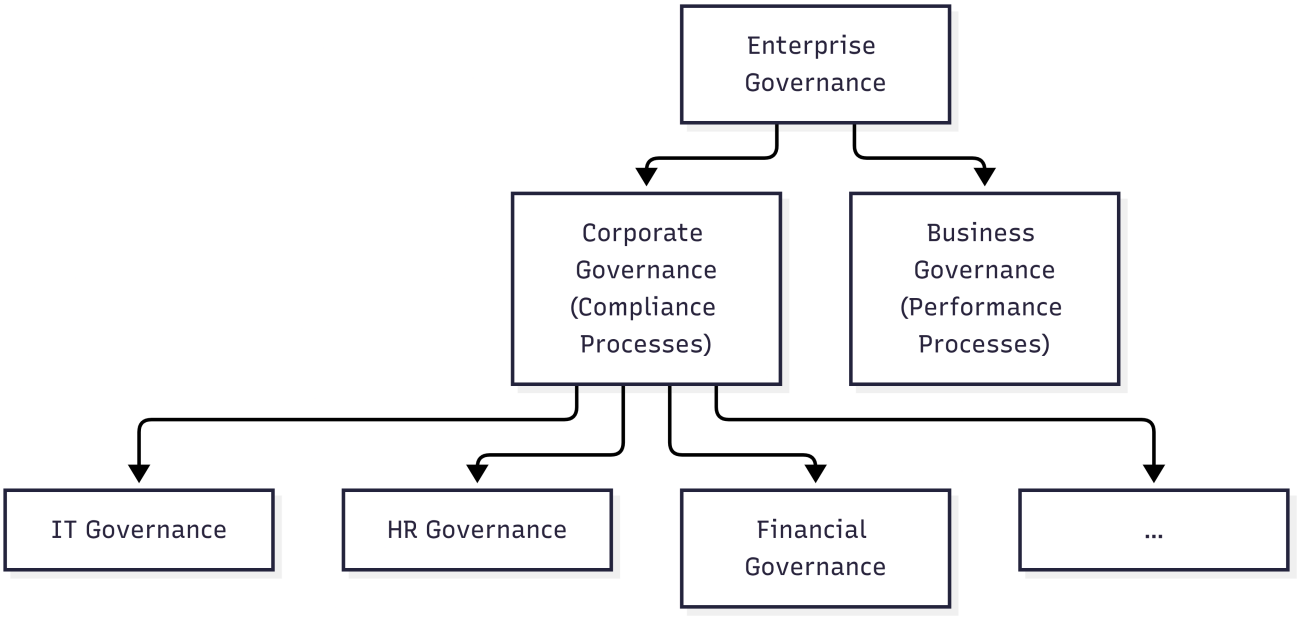
If the direct/control loop is to be steered intelligently, it must be steered against a measurable quantity. That quantity is **risk**, which may take financial, production-related, cybersecurity, or other forms. The IT Governance Institute captured this centrality when it redefined enterprise governance as the set of responsibilities and practices exercised by the board and executive management with the aim of providing strategic direction, ensuring objectives are achieved, ascertaining that risks are managed appropriately, and verifying that the enterprise's resources are used responsibly.

Once risk is accepted as the governing metric, the entire governance problem collapses into a clean **optimization process**. The direct half of the loop is charged with a single objective: reduce risk, by putting in place actions that lower the risk level. The control half of the loop is charged with the complementary objective: measure the reduction, by taking measurements, recomputing the risk, and aggregating it into a coherent picture. Governance thus becomes the continual minimization of risk under observation.

Decomposing Governance: IT Governance and Assets

Supporting business and corporate governance in practice requires more specialized sub-governances, each dedicated to a particular domain of the enterprise. Enterprise governance decomposes into corporate governance and business governance at the top, and beneath these sit the domain-specific

governance sub-systems: **IT governance**, **HR governance**, **financial governance**, and others. The following diagram shows this decomposition.



IT governance is defined as the set of processes that ensure the effective and efficient use of IT in enabling an organization to achieve its goals. It begins from the infrastructures and examines every kind of risk that arises from the digital environment, and especially the risks that arise from the *offering of services*. IT-based systems create serious risks precisely because they manage and store the majority — if not the entirety — of a company's electronic assets: the data and information stored inside the organization, the data and information transmitted across its networks, and all the systems and applications required to store, transmit, and process that data. These electronic assets are exposed to a wide range of threats aimed at compromising their three cardinal properties, **Confidentiality, Integrity, and Availability**, collectively abbreviated as **CIA**.

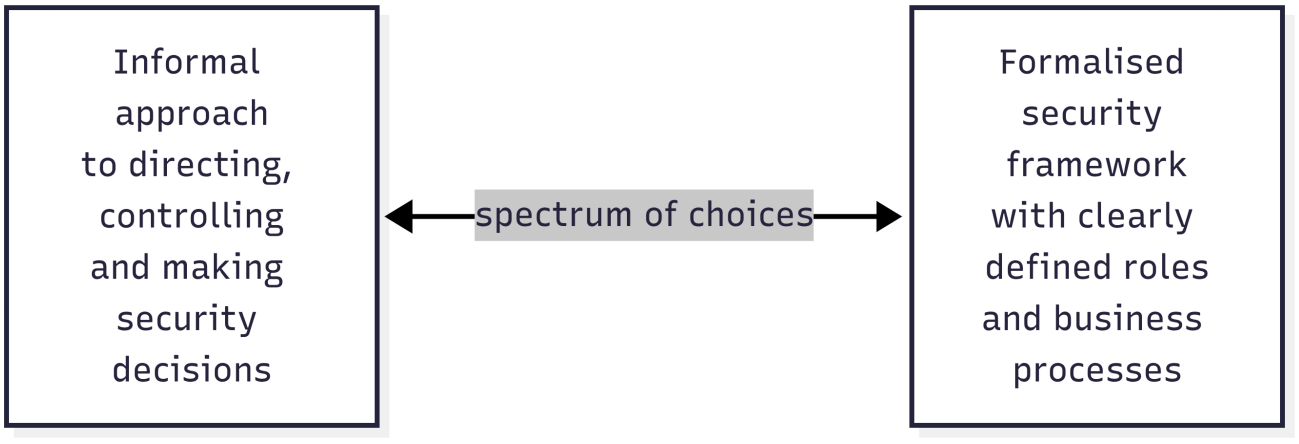
The decisive insight of IT governance is that the infrastructural view must be fused with the business view, because the two views assign radically different importance to physically identical machines. Consider a network of interconnected servers. On one server sits an archive of data; on another, identical machine sits the web server that hosts the company's online shopping site. Viewed through the infrastructure lens alone, the two servers look equivalent — both are connected, both are running, both are simply nodes in the topology. The moment the business lens is applied, the equivalence collapses. If an attacker brings down the shopping site, the company can no longer serve its customers and immediately begins to lose money, which is a serious risk. If an attacker damages the archive, the data can very often be restored from other sources, which is a comparatively minor risk. The two machines are the same; the risk they carry is not. This is why IT

governance insists on considering *all* aspects of the digital world — not merely the machines, but everything operating around them — and on first identifying which assets genuinely matter. Running a live service is an asset in the critical sense; the archive, in this example, is not an asset of the same order.

The threat landscape, moreover, is not confined to the outside. One must protect against external attackers arriving from the Internet, but one must equally guard against attackers already *inside* the organization. Employees who have become unhappy are a particularly grave danger: they know the assets intimately, they understand how the infrastructure is managed, and this superior knowledge makes them far more dangerous than an outsider who must first discover what an insider already knows.

Choosing an Approach: The Spectrum of Formalization

Having established that governance is a risk-driven control loop operating over identified assets, the practical question becomes how to implement that loop in a concrete company. There is no single universal answer — there is **no "one size fits all" approach** to security governance. Instead there is a spectrum. At one extreme lies a fully **formalized security framework**, with clearly defined roles and rigorously defined business processes. At the opposite extreme lies an **informal approach** to directing, controlling, and making security decisions. Every real company must find its proper place somewhere along this continuum.



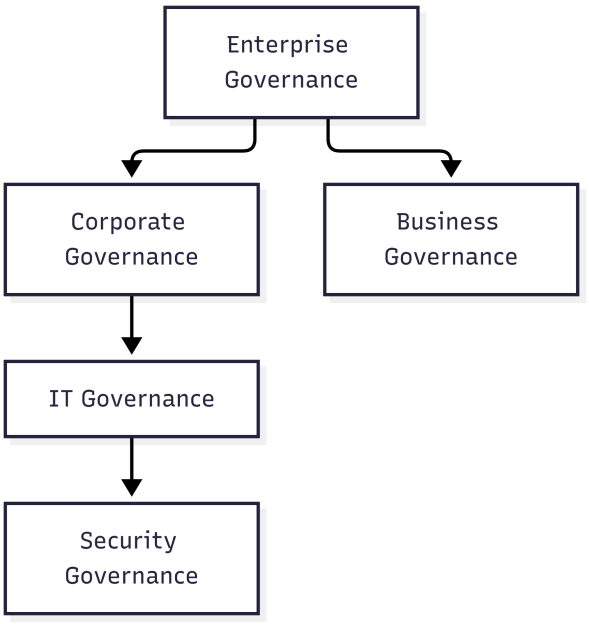
Four questions guide a company toward the right level of formalization. The first concerns **size and complexity**: how large and how complex is the organization? Small companies must lean on flexibility because they typically lack the resources for heavy formalism, while large companies will most probably require a formalized security framework simply to keep coordination tractable. The second concerns **resources**: what budget is actually available for security governance? A formalized

approach is expensive — it demands certifications and sustained investment — so a company without money cannot realistically adopt it. The third concerns **mission and the importance of security to it**: what does the organization do, and how important is security to those aims? For a bank, security is a top priority and a matter of existential dependence; for a manufacturer, security still matters but the business does not hang on the same human-facing service dependencies. The fourth concerns **external considerations**: are there contractual, legal, regulatory, or sector-specific requirements pushing the company toward a particular level of formalism? A small company might in itself prefer an informal posture, yet find itself obliged to obtain certification because it works for a public organization that demands it.

Answering these questions well requires analysing the **context**, and in practical terms the correct approach comes down to identifying three things. One must identify the **what**, meaning the security decisions to be made. One must identify the **who**, meaning the deciders who will make them. And one must identify the **data**, meaning the information required to make sensible and informed choices. These three together — decisions, deciders, and the information feeding them — constitute what is meant by a **security framework**.

Positioning and Purpose of Security Governance

Security governance can now be located precisely within the larger structure. It is the means by which an organization controls and directs its approach to security, and when it is done well it coordinates all of the organization's security activities and enables the flow of security information and security decisions throughout the organization. Structurally, security governance is a **vertical** concern that cuts across the other governances: it sits within IT governance, which sits within corporate governance, which — alongside business governance — sits within enterprise governance. The nesting looks as follows.



The reasons to care about security governance follow directly from this position. It provides a structured approach to managing and mitigating cybersecurity risks within the organization. It helps protect sensitive data, maintain regulatory compliance, and strengthen the trust and confidence of stakeholders. And by implementing effective security governance, an organization reduces both the likelihood and the impact of security incidents, thereby safeguarding its reputation and its business operations.

This defines the arc of the entire subject. The study of security governance will proceed through models for information security governance and the vertical nature of security; through cybersecurity best practices, guidelines, and frameworks and how to use them; through the assessment and management of risk, which entails understanding threats, vulnerabilities, and attacks, evaluating their impact, and finding proper countermeasures; through the modelling and understanding of threats; through the management of incidents; and finally through legislation, compliance, and insurance, closing on open research issues in security governance.

A Model for Information Security Governance

Recap: What Information Security Governance Is

Before assembling the full model, it is worth restating the object of study with precision. **Information Security Governance (ISG)** is the process of establishing and maintaining a framework, together with the supporting management structure and processes, that provides assurance of several things at

once: that information security strategies are aligned with and support the business objectives; that they are consistent with applicable laws and regulations through adherence to policies and internal controls; and that responsibility is properly assigned throughout the organization — all of this undertaken in an effort to manage risk. Said more plainly, ISG is the global effort a company makes to ensure the well-being of its electronic resources.

The Principle of Cost-Effectiveness

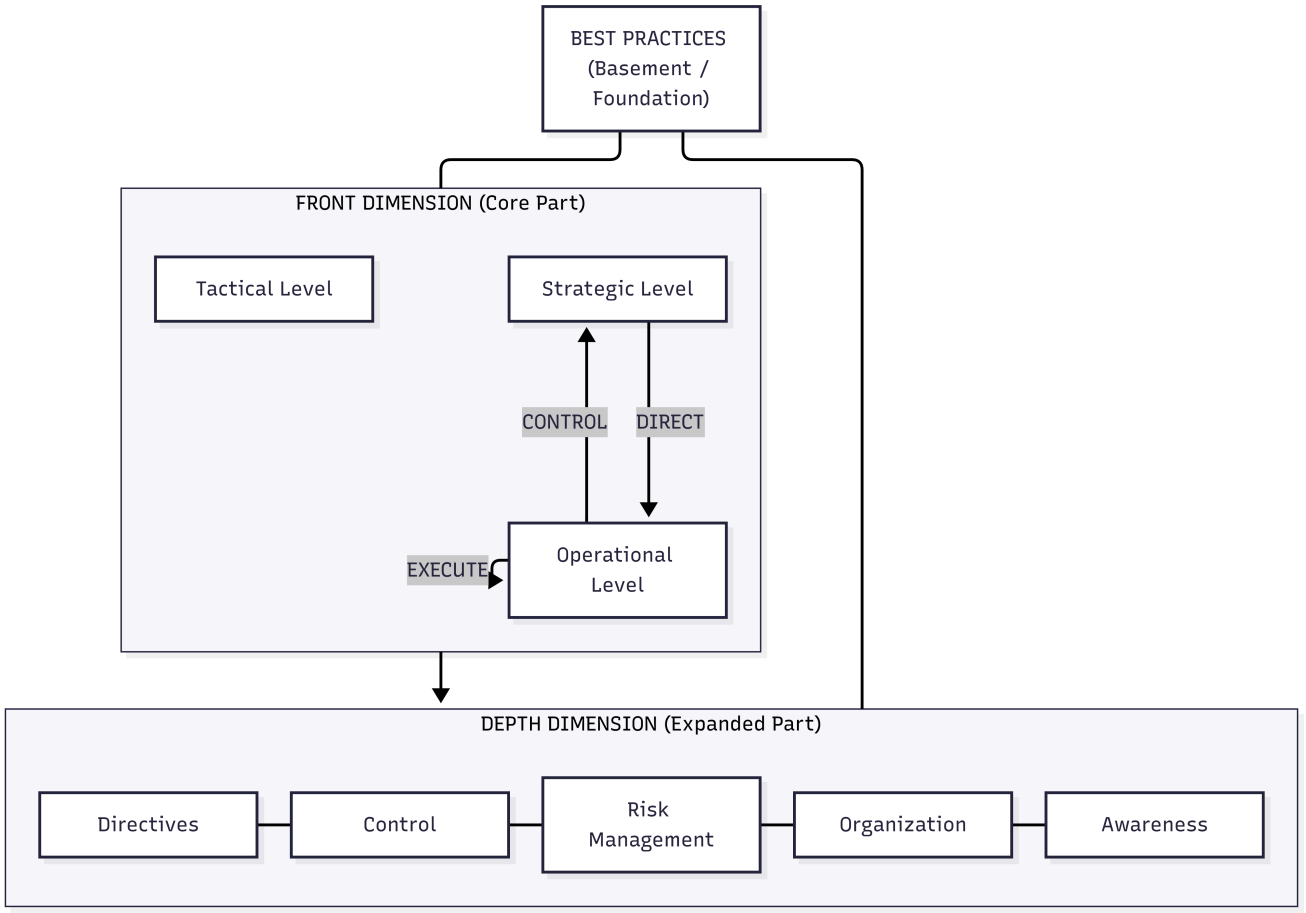
The first foundational principle on which any ISG framework must be built is that it has to be **cost-effective**. Security is expensive, and there must therefore be a deliberate balance between the cost of protecting electronic resources and the risk to which those resources are exposed. This balance has two failure modes, each equally to be avoided. **Overprotection** wastes money on defences whose cost exceeds the harm they prevent, generating unnecessary expense. **Underprotection** leaves genuine exposures unaddressed, allowing risks to materialize and strike the company. Good governance lives in the narrow band between these two errors.

The practical force of this principle becomes visible the moment one imagines the two possible worlds. Suppose a risk assessment has been carried out and has produced a list of risks, each tagged with an associated security level. In an idealized world with no budget limit, one could simply march down the list and remediate every risk one by one until none remained. Real organizations do not live in that world: they operate under a **fixed budget** and must decide where that budget goes, starting from the most critical risks and working downward. A characteristic situation then arises in which most of the budget has been consumed fixing the critical problems, and a residual amount remains. At this juncture the decision must be made carefully, weighing the cost of fixing a given remaining risk against the cost that would be incurred if that risk actually materialized — neither overspending on marginal exposures nor abandoning issues that quietly matter. Even the act of *identifying* a risk depends on acquiring and analysing data with the right instruments; if the proper instruments are not used, important points may simply be missed, and the balance will be struck on incomplete information. For all these reasons, a good **IT risk management strategy** is mandatory as the precondition for a good ISG strategy, and good governance as a whole demands a strategy that weighs multiple aspects rather than fixating on the infrastructure alone.

The Von Solms Model: Overview

The model used to organize all of these elements was introduced by **R. Von Solms in 2006**. It can be pictured as a solid three-dimensional wedge whose purpose is to gather every element one must keep in mind and to arrange those elements in an interconnected structure. The wedge has three distinguishable parts, and understanding the model means understanding what each part contributes.

The **Front Dimension**, also called the **Core Part**, is the triangular face of the wedge. It is the implementation of the control loop already studied, spanning the three management layers of the pyramid. The **Depth Dimension**, also called the **Expanded Part**, is the set of five layers extending backward from that front face: **Directives**, **Control**, **Risk Management**, **Organization**, and **Awareness**. Finally, the **basement** on which the whole wedge rests is the **Best Practices** layer, the foundation from which everything else is constructed. Best practices are the documents that collect the common, accumulated knowledge of the field in terms of cybersecurity and management. The overall structure is shown below.



The Front Dimension (Core Part)

The Front Dimension represents the execution of processes and actions and the influence of the Direct and Control loop upon those processes. It rests on two core principles.

Core Principle 1 states that the model covers the three well-known levels of management — strategic, tactical, and operational. This principle answers the question of *why all three levels must be covered by the governance process*: every level possesses its own distinct view of the company, and no single level can see the whole. At the top, a handful of people take general decisions and, when the direct process begins, they possess only ideas about what to do; the directive documents they eventually write are therefore quite general. At the bottom, people are in direct contact with the system, laying their hands on machines and producing procedures, and their scope is extremely detailed and correspondingly restricted. Because each person's view is limited in this way, the intermediate tactical layer is indispensable: its whole function is to translate the general into the detailed on the way down, and, on the way up, to translate the detailed back into the general. This is why the processes must be able to flow *vertically* in both directions. On the control path, the operational level begins by collecting information of great precision — numbers of alerts, flows across different machines, and similar indicators — and this precise information is progressively correlated with the procedures and abstracted as it climbs, so that the movement upward is a movement from a very detailed view toward an increasingly summarized one.

Core Principle 2 states that across these three levels there are three very distinct *actions*, and that control at the bottom is emphatically not the same activity as control at the top. The rationale is easy to appreciate from the Board's perspective: the Board is not interested in learning that 321 virus attacks occurred last week, that 27 software patches were applied, and that 7 anti-virus updates took place — this is far too much detail to plan with. What the Board wants to know is whether the company's access to the Internet causes any *unmitigated* risks to the company. If there are none, the Board is content; if there are some, the Board wants to be told precisely which unmitigated risks these are. The action of control must therefore be qualitatively different at each altitude.

The Direct Part, Layer by Layer

The downward Direct process can be read as a matrix in which each management layer has a distinct meaning, a specific input, and a specific output, and in which the output of one layer becomes the input of the next.

At the **strategic level**, to direct means to **identify the assets, their relevance, and their required level of protection** — appropriate here because the strategic level alone possesses the overview of the whole company and can therefore say what must be protected and to what extent. Its **input** is the

context, split into internal and external factors. The internal factors concern the company's own world: its strategic vision, the role of IT, its competitiveness, its mission, its constraints, and its production processes. The external factors concern the surrounding world: legal and regulatory prescriptions and other external risks. Its **output** is a set of **Directives** indicating, at a high level, what the Board expects to be done as far as the protection of the company's information assets is concerned.

At the **tactical level**, the directives received from above are *expanded* into sets of relevant information security **policies, company standards, and procedures**. Here the macro-objective is analysed against the situation and translated into middle-level objectives. The **input** is the set of directives produced above, and the **output** is the collection of policies, procedures, and standards with which the company must comply — the way digital resources are managed, the way accounts are handled, and so on.

At the **operational level**, the inputs are expanded further into sets of **administrative guidelines and administrative procedures**, and the technical measures are physically implemented and managed. The **output** is the set of **operating procedures** specifying exactly how things must be done, and this output forms the very basis of execution on the lowest level.

The Control Part, Layer by Layer

The upward Control process is governed by a single, decisive maxim: **you can only manage what you can measure**. To control properly one must measure; to measure one must know in advance which information and data to collect. This *measurability* characteristic cannot be an afterthought — it must sit at the centre of all directives, policies, standards, and procedures produced during the Direct part of the model, so that the objects being directed are constructed to be measurable in the first place. The Control matrix mirrors the Direct matrix but runs in the opposite direction, from operational up to strategic.

At the **operational level**, to control means that **measurement data is extracted from a wide range of entities**, either automatically or manually. Concretely, this means deploying monitors — **agents** — into the environment to collect and extract data. These agents may live on the machines themselves or in the network, and their power depends on their observation range. Here a fundamental tension must be respected: **the need for monitoring must not compete with the needs of the business**. Observing everything by placing

agents everywhere is expensive and degrades the system's speed; when one interferes with the system to monitor it, one cannot do too much, or the system will suffer. The guiding rule is that the need for protection must exist and survive alongside the need of the business, without smothering it. The **input** at this level is the raw **measurements**, and the **output** is a set of **specialized reports** created on this level from the extracted operational data, which is then passed upward to the tactical level.

At the **tactical level**, control means **measurement and monitoring against the requirements of the relevant policies, procedures, and standards**. The tactical layer receives all the operational reports, aggregates them together, and attempts to identify existing threats, evaluating whether the policies in force are sufficient. Its **output** is a set of **tactical management reports** indicating the levels of compliance and conformance, which are then encapsulated and passed to the strategic level.

At the **strategic level**, control culminates in **Situational Awareness**. The **output** at this apex consists of **reports reflecting compliance and conformance to the relevant directives, including risk considerations** — the very snapshot the Board needs in order to decide whether to hold course or re-direct.

The Basement: Best Practices

Underneath both the Front and the Depth dimensions lies the layer of **best practices**, which sets the ground on which all operations are instantiated. Best practices are documents collecting the knowledge that people have acquired over the years in terms of security and management. The specific best practice one reaches for depends on the specific problem to be solved, and the required granularity depends on the management level being served.

The purpose of the whole ISG model, phrased memorably, is to help the organization in *"doing the right things right."* This slogan resolves into two questions that every **Information Security Manager (ISM)** must be able to answer. First: *how do I know what the right things are?* Second, supposing that is known: *how do I know I am doing them right?* The answer to both is: look to best practices. Formally, best practices — also called standards or guidelines — are a set of documents reporting the experiences and solutions arrived at by experts in information security management. They can be understood as the consensus of experts in the field, and they provide an internationally accepted framework usable

as a building block for ISG. Many of them exist, each focused on related but different aspects, including the **ISO 27000 family**, **COBIT**, the **NIST SP 800-*** series, the **CSC** (Critical Security Controls), and the **NIST CSF** (Cybersecurity Framework), among others.

ISO 27002

Among these best practices, **ISO 27002** deserves close attention because companies use it to set up their infrastructures from the policy point of view. It is an International Standard designed to be used by organizations that intend to do any of three things: to select controls within the process of implementing an Information Security Management System (ISMS) based on ISO/IEC 27001; to implement commonly accepted information security controls; or to develop their own information security management guidelines.

ISO 27002 is fundamentally a collection of **controls** — points that need to be verified. In its 2013 edition it contains **14 security control clauses**, which collectively hold a total of **35 main security categories** and **114 controls**. (These controls are general security controls and are not themselves objects of certification; they function as instructions rather than as a pass/fail certificate.) The structure is strictly hierarchical: each clause that defines security controls contains one or more main security categories, and each main category in turn contains two things — first, a **control objective** stating what needs to be achieved, and second, one or more **controls** that can be applied to achieve that objective.

The description of each individual control is itself standardized into a fixed structure of three parts. The first part is the **Control** proper, which defines the specific control statement needed to satisfy the control objective. The second part is the **Implementation Guidance**, a longer text that provides more detailed information to support the implementation of the control and the meeting of its objective; this guidance carries an explicit caveat, namely that it may not be entirely suitable or sufficient in every situation and may not fulfil an organization's specific control requirements. The third part is **Other Information**, which provides further considerations such as legal factors and references to other standards, and which is simply omitted when there is nothing further to say. In this way a control functions as a security checklist item that matches the requirements for certification, accompanied by a guideline explaining one possible way to implement something compliant with the check.

ISO 27001 and the Distinction Between the Two Standards

ISO 27001 is the companion International Standard that specifies the requirements for establishing, implementing, maintaining, and continually improving an ISMS within the context of the organization. It also includes requirements for the assessment and treatment of information security risks tailored to the organization's needs, and it is deliberately generic, intended to be applicable to all organizations regardless of type, size, or nature. Crucially, ISO 27001 is the standard that grants **certification**: it certifies that a system is compliant with its requirements, and it operates as a checklist of security checks that a company must satisfy to be formally certified.

The relationship between the two standards is best understood as a division of labour, and the difference between them is the difference between a requirement and a guideline. **ISO 27001** is very specific and strict; it spells out in detail what a company must comply with and must have in place in order to be formally certified. **ISO 27002**, by contrast, is a *guideline* document. It advises companies on what they should have in place as far as their information security management is concerned in order to follow best practice, guiding a company to structure its information security management according to the accumulated experience of other companies. It is deliberately rather high-level and does not drill down into very detailed specifics. The compact way to remember the pairing is that **27001 certifies and is strict, while 27002 guides and is advisory**.

The Directive and the Information Security Policy Architecture (ISPA)

Why a Documented Direct Process Is Needed

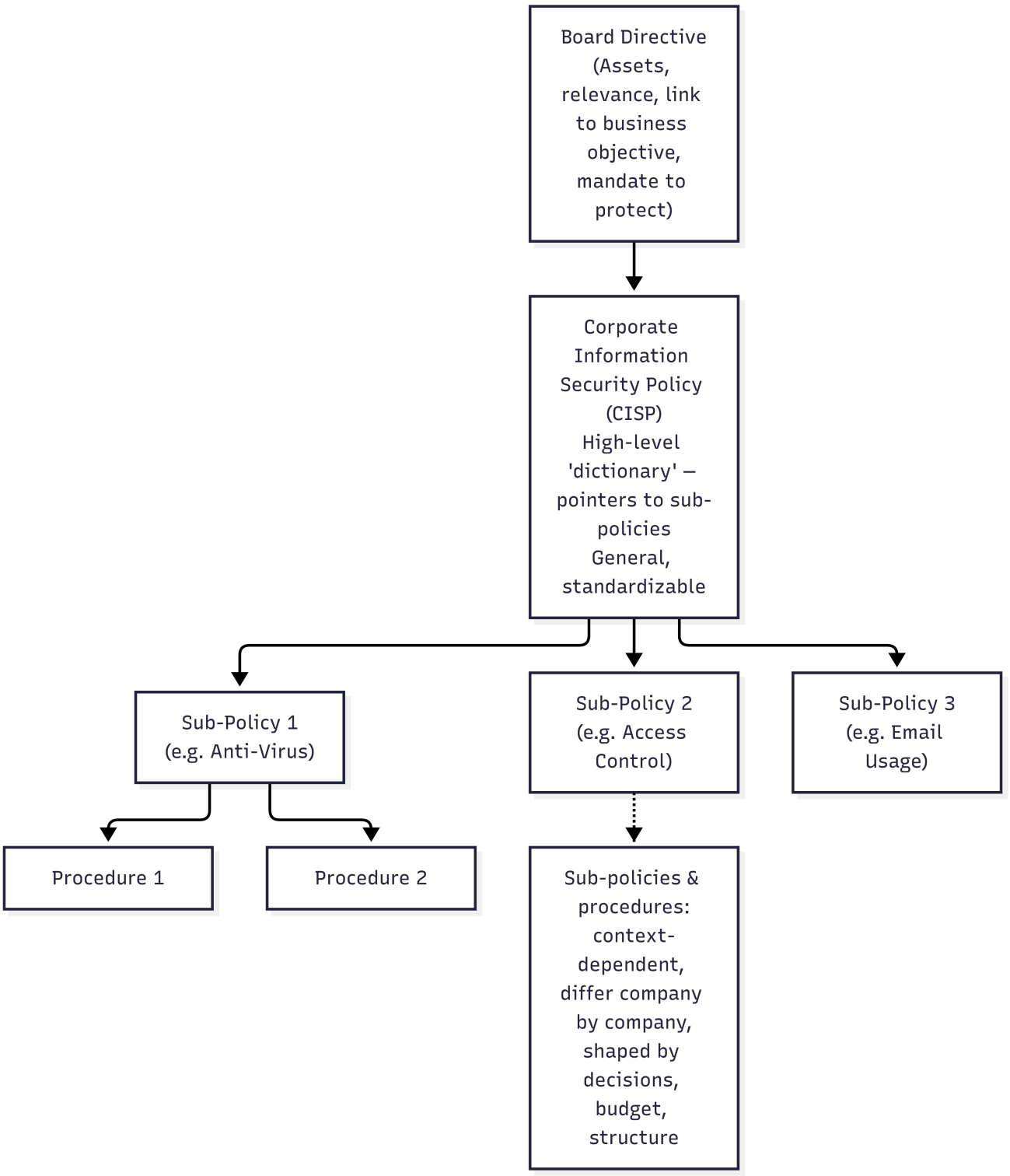
Every best practice contains, in some form, the notion of a **Directive**, and the task it imposes is concrete: to set up the documents and policies that dictate how security is to be governed inside a company. Chapter 5 of ISO 27002, for instance, requires that decisions be taken and acted upon in a *structured* way, which in effect demands the design of a process that supports all three management layers. To meet that demand, an organization must develop a **framework** — a methodological instrument, a defined set of things that must be done — capable of supporting the Direct process. This framework is not a single document but an

interconnected architecture of documents, each linked to the others, which together implement the guidelines of the downward direct flow.

The reason everything must be written down is that documentation is what confers **trust and value**. By formalizing rules and policies on paper, the organization gives weight and accountability to what it does; an undocumented intention carries no assurance, whereas a documented one can be communicated, enforced, audited, and relied upon. Because the Direct process flows from top to bottom, and because each layer operates at a different degree of abstraction, the architecture must provide a set of documents calibrated to the right level of information at each layer. A general document is produced at the top and is connected downward to progressively more specific documents produced in the layers beneath it.

The Document Tree

The architecture of the Direct process is best visualized as a **tree**. At the root sits the **Board Directive**. One level below sits the **Corporate Information Security Policy (CISP)**, which receives the directives from the board and transforms them into guidelines for the implementation of security controls. From the CISP, a set of **sub-policies** branches out, one for each distinct domain of concern — access control, human resource management, device management, and so on. From those sub-policies are derived the **company standards**, which express the minimum level of specification that must be achieved, and finally, at the operational layer, the **procedures**, which are protocols: ordered lists of concrete actions. The flow and its contents are shown below.



A crucial distinction runs through this tree. The **Board Directive** and the **CISP** are relatively **general and standardizable**, because almost every company shares essentially the same fundamental need for protection, and so the upper two documents can look broadly similar across organizations. The **sub-policies** and the **procedures**, by contrast, are strictly **context-dependent**: they vary from company to company and are shaped by the particular decisions the organization wishes to enforce, by its structure, and by the budget it has available.

The Board Directive

At the top of the tree, the Board Directive carries the raw material of the entire Direct process. It must contain two things above all: the **set of IT assets identified by the Board**, together with their relevance and their connection to the business objective, and the **mandate from the Board to protect those assets**. The inclusion of relevance is essential, because protection must proceed *asset by asset* — not every asset carries the same importance, and the directive must make those differences explicit so that protection effort can be allocated accordingly.

The Corporate Information Security Policy (CISP)

One level lower sits the CISP, the **high-level document that provides the basis for all lower-level documents related to information security**. Its function is to take the objective expressed in the Board Directive and translate it into a middle-layer objective, and from that translation it extends a set of pointers aimed at the specific sub-policy documents below. In this sense the CISP operates as a kind of **dictionary**: it allows the reader to interpret and map the sub-policies against the original board directive, binding the detailed and the general together.

The methodology offers a set of guidelines for writing a proper CISP, specifying both what it should contain and how it should be formalized. There are ten such guidelines.

First, the CISP must indicate **board and executive management support and commitment**, and it must be clear that the CISP flows from a higher-level directive; the link back to the Board Directive and the enforcement behind it must be visible. Second, it must be **accepted and signed by the CEO or an equivalent officer** — the signature of a board member is what gives the document its authority. Third, it **must not be a long document, nor written in technical form**: a maximum length of about four to five pages is appropriate, and it should contain high-level statements concerning information security rather than implementation detail. Fourth, it **should not change very often and must be stable** with respect to technical developments and changes; because it is an intermediate working document bridging the direction and the operational levels, frequent revision would destabilize everything hanging beneath it. Fifth, and as a direct consequence of the stability requirement, the CISP **must not contain any references to specific technologies and must be technology-neutral**, since technology evolves far too frequently to be enshrined in a stable document.

Sixth, the CISP must indicate **who owns the policy** and what the responsibilities of other relevant people are; stating roles and responsibilities is what creates a structure that links each action to a specific person, producing a map of who is supposed to do what and who is supposed to monitor it. Seventh, it must clearly indicate the **scope** of the policy — that is, all the people who will be subject to it. Eighth, it must refer to possible **disciplinary actions for non-conformance**, both to the CISP itself and to its lower-level constituent policies; these disincentives are necessary precisely because of the human factor, since people tend to look for an escape strategy when confronted with something difficult. Ninth, it must be **distributed as widely as possible** throughout the company and covered in all relevant awareness courses, so that employees understand both the policies and the reasons they exist. Tenth, it must contain a **Compliance Clause**, the element that ties the document directly into the Control process.

A representative CISP is organized into clearly named but deliberately general sections — the importance of information to the company, what information security is, why the company needs it, the information security commitment of the board, the scope, the information security statements that make up the policy, the responsibilities, and the compliance clause. The document is short, and its section titles are generic. Where it does become concrete is in the **scope**, which must define exactly who is affected by the policy: typically all of the company's IT assets stored, processed, and distributed via its information processing systems; any person granted authorization to access those resources, including permanent, temporary, third-party, and contractual employees and users; and all business partners and clients that access the company's IT assets in any way.

Sub-Policies

Descending one further level, the **sub-policies** take the mandate to protect the assets — already translated into macro-objectives by the CISP — and define specific policies for the specific aspects that must be considered. These documents are **company-dependent**: each organization must define them according to its own structure and circumstances. Nevertheless, certain categories of sub-policy apply to essentially everybody and are commonly defined by all organizations: a **Malicious Software Control Policy** (an anti-virus policy), an **Acceptable Internet Usage Policy**, an **Acceptable Email Usage Policy**, a **Logical Access Control Policy**, a **Disaster Recovery (Backup) Policy**, a **Remote Access Control Policy**, and a **Third Party Access Control Policy**.

One requirement overrides all others at this level and is marked as critically important: the ISPA must **clearly indicate where each of these different policies originates**. In concrete terms, every sub-policy must be *linked* or *traced back* to the CISP in some way, so that the chain of authority from the board directive down to the individual policy is never broken. A well-formed sub-policy therefore carries an explicit **Reference** section pointing *up* to the statements in the CISP that give it origin, and *down* to the lower-level procedure that supports it, along with a **Main Policy Statement** and its supporting sub-statements. For example, an Internet Acceptable Use sub-policy would state that use of the company's Internet service must be in support of official company work, that all user requests for connectivity must be approved and supported by the employee's line manager and made through a specified application procedure, and that it is the line manager's responsibility to ensure the user does not misuse the service.

Procedures

At the bottom of the tree lies the **operational level**, inhabited by the **procedures**, which are effectively **algorithmic steps**. Where a sub-policy declares what must be true, a procedure specifies the exact ordered actions that make it true. In the running example, a user requesting Internet connectivity within the company triggers an application procedure that states precisely what must be done: the applicant records identifying details, signs a declaration that they have read both the CISP and the relevant Internet Acceptable Use sub-policy and will act in line with them, and obtains the signatures of the line manager and the information security officer. A procedure of this kind must also **clearly state what it means to be compliant with it**, carrying its own compliance clause so that conformance can later be checked.

Once the full set of documents is in place — stating, from top to bottom, what the organization needs to do — the procedures are what get implemented in the **Execution** part of the model. Execution is simply the carrying-out of those procedures, and the role of the Control process that follows is to verify that everything is in fact working as the documents require.

The Control Part and Compliance

Compliance as the Scope of Control

Once the documents of the Direct process exist and the procedures have been executed, the model turns upward into the **Control** part, whose main scope is the

compliance check. ISO 27002 makes this requirement explicit: after everything has been put in place, the organization must verify compliance at the various levels — compliance with external laws and contractual requirements, and compliance with the security policies and standards it has established internally. The standard devotes an entire clause to this, titled **Compliance**, with the objective of avoiding breaches of legal, statutory, regulatory, or contractual obligations related to information security and of any security requirements.

Two of its controls capture the essence of the task. The control on the **identification of applicable legislation and contractual requirements** demands that all relevant legislative, statutory, regulatory, and contractual requirements — and the organization's approach to meeting them — be explicitly identified, documented, and kept up to date for each information system and for the organization as a whole, with the specific controls and individual responsibilities also defined. The control on **compliance with security policies and standards** demands that managers regularly review the compliance of information processing and procedures within their area of responsibility against the appropriate security policies, standards, and other security requirements. ISO 27002 further requires that management initiate an **independent review**, carried out by individuals independent of the area under review — the internal audit function, an independent manager, or an external specialist — to ensure the continuing suitability, adequacy, and effectiveness of the organization's approach to managing information security; the results are recorded and reported back to the management that initiated the review, and if the approach is found inadequate, corrective actions must be considered.

The Measurability Problem

The central difficulty of the Control part is defining the **compliance criteria** — that is, deciding how to verify that the organization is in fact behaving correctly. This difficulty flows directly from the nature of the documents themselves. Policies, directives, and standards are written in **natural language**, and natural language is neither strict nor formalized; consequently it is genuinely hard to establish exactly what must be verified. The problem reduces to the governing maxim of the whole control process: **you can assess only what you can measure**. Things that cannot be measured cannot be controlled, so the organization must define measurements that make it possible to quantify how compliant it is. For this reason the compliance clauses, which encode these measurements, must provide a clear metric that can be evaluated and assessed.

Evaluating compliance is not equally hard at every level. At the **bottom** of the pyramid it is comparatively easy, because the operational layer deals in procedures and in concrete data drawn directly from the system, which can be monitored. If an access control policy declares that only authorized users may access a certain type of data, compliance can be checked simply by inspecting the system's logs. As one climbs **upward**, however, the criteria become far more general, and the natural question arises of what metric could possibly apply at those altitudes. The answer is the concept of **risk**. At the higher levels, verifying compliance means **assessing the risk that derives from non-compliance** with the policies and directives. The task, therefore, is to translate the technical data extracted from the environment into something that allows the risk level to be evaluated — to convert log entries and alert counts into a statement about exposure.

The Role of Time

In carrying out this exercise, **time plays a decisive and troublesome role**. Assessing risk is expensive in time for several compounding reasons. First, the necessary information is widely spread throughout the company and must be gathered from many places. Second, that data must then be analysed, which itself consumes time even when software is used to assist. Third, a conclusion must be derived from the analysis. Taken together, a full risk assessment can take **weeks or even months**. It cannot be performed daily; it is something that must be planned.

This temporal cost creates a structural tension with the domain itself. In cybersecurity, **weeks constitute a very large temporal frame** — the threat environment can change completely within such a window. Worse, the world is **dynamic**, so even if the organization manages to choose what seems like the right assessment interval, the context captured at the start of the assessment may no longer hold by the time the assessment concludes. Traditionally, compliance monitoring has been performed through periodic **ICT audit sessions**, in which ICT risks are evaluated, an audit report is produced, and its results are compared against the compliance clauses. The abiding issue with this approach is precisely timing: the world is dynamic, and monitoring ought ideally to be done in near real time, which the periodic-audit model struggles to deliver.

Guidelines for Writing Compliance Clauses

Unlike the Direct process, which comes equipped with templates and suggestions for writing the CISP and sub-policies, the Control process offers **no standardized**

content for compliance clauses — specifying them is not an exact science, and very few guidelines are available. Only a small number of principles can be stated. A compliance clause must be **clear and precise**, deliberately avoiding the insertion of ambiguity; otherwise two people performing the same assessment against the same clause might reach different conclusions simply because the criteria were open to interpretation. A clause such as *"Employees must take Information Security seriously"* is a textbook failure, because the word *seriously* has no fixed meaning and cannot be assessed. Beyond clarity, a compliance clause should **express a way to measure the degree to which it is satisfied**, providing a usable metric rather than a vague aspiration.

Worked Example — A Clause for the Board Directive

Consider an extract from a Board Directive in which the board accepts the importance of the company's IT infrastructure to its strategic goals, recognizes the potential impact a compromise could have, and consequently demands a proper environment to protect the IT infrastructure against any disruptions or compromises, fully supporting a proper ISG environment. Here the **asset** is the IT infrastructure and the **protection need** is protection against compromises. Two candidate compliance clauses illustrate the difference between a weak and a strong formulation.

Option 1 states that the board requires proper feedback, every three months, of the level of compliance with the directive. Its merit is the presence of a clear **time window**. Its defect is that the phrase *level of compliance* is never defined anywhere — it is left ambiguous exactly what is to be assessed. **Option 2** states that the board requires detailed feedback, every three months, of any serious risks that may compromise the protection of the company's ICT infrastructure. This option retains the same time window but is markedly better, because it also clarifies the **metric**: the thing to be reported is *risk*. Option 2 is therefore the superior clause.

Worked Example — Clauses for the CISP

The second example concerns the CISP, which, as established, is a document articulated as a collection of statements that act as pointers to sub-policies. Because of this structure, it is preferable to attach a compliance clause to **each statement** (or each category) rather than to the document as a whole. Taking Statement 6.1 — which declares that the company will have a proper information

security organizational structure to manage information security, ensuring that security roles are assigned, that users are aware of the policy and of the disciplinary consequences of non-compliance, and that implementation is continuously coordinated and reviewed — two candidate clauses can again be compared.

Option 1 states that the effectiveness of the company's information security organizational structure in managing and enforcing the requirements of the ISPA shall be evaluated every six months, with the results incorporated into higher-level reporting. **Option 2** states that the way the information security function is organized, and the reporting structures within that function, shall be evaluated every twelve months to ensure the structure remains adequate to protect all the company's electronic assets, with the results forming part of the annual evaluation and reporting of organizational structures. Both options share the strength of a defined **time frame**. Option 1's weakness is that it does not specify *what* must be reported, so a change in the person writing the report would produce a different view each time. Option 2 improves on this because it refers to a document that is already expected to be produced annually and will therefore most probably be structured in a consistent way — yet it, too, leaves the **metric** unexpressed, failing to say explicitly what is to be measured. The honest conclusion is that **no option here is perfect**, but Option 2 is better than Option 1. The recurring lesson across both examples is that a good clause pins down both a time window and a measurable metric, and that most real clauses succeed on one dimension while remaining imperfect on the other.

Risk Management

Risk as the Element That Drives Control

From the control point of view, the single element that makes the control loop implementable is **risk**. Risk, however, is perceived differently at every layer of the hierarchy, and to deal with it coherently the organization needs a dedicated process for its analysis: **Risk Management**. This process must be **vertical**, assessing risk at every level, and the closer it operates to the bottom of the pyramid, the more tightly it is bound to the technological layer.

Formally, risk management is the process of identifying and assessing all potential risks, as well as introducing the controls that should mitigate those risks to acceptably low levels. It is one of the security processes required to identify the risks that may impact the well-being of an organization. Its defining spirit is to

begin by looking for what might be dangerous, to establish to what extent it is dangerous, and then to seek possible solutions evaluated against a **cost–benefit principle** — the same cost-effectiveness discipline that governs the whole of ISG.

The Two Dimensions of Risk

Although there are several ways to perform the process, risk is generally understood through **two orthogonal dimensions**. The first is the **probability or frequency** — how likely it is that something bad will happen. The second is the **magnitude of gains or losses**, usually called the **impact** — how severe the consequence would be if it did happen. The aim of risk management is accordingly to determine two things: what the impact will be if the risk materializes, and how often the risk might materialize. These two dimensions combine into the canonical formula:

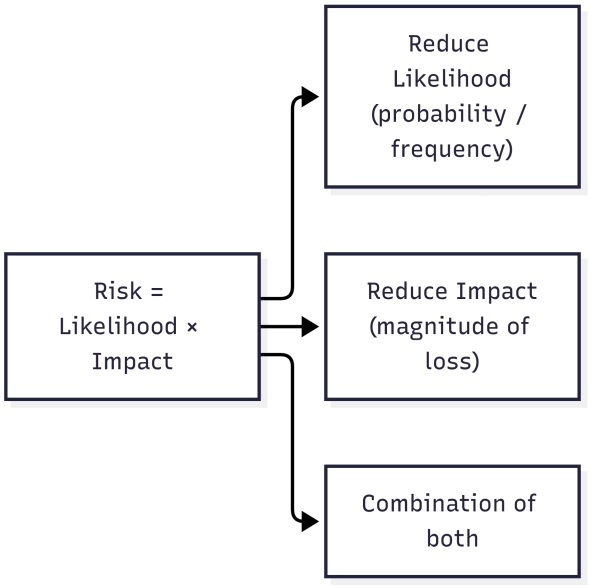
$$\text{Risk} = \text{Likelihood} \times \text{Impact} = (\text{Threat} \times \text{Probability}) \times \text{Impact}$$

The relationship is often displayed as a coloured matrix with likelihood on the vertical axis and impact on the horizontal axis, in which the combinations shade from green in the low–low corner, through amber in the middle band, to red where high likelihood meets high impact — a visual map that lets managers read off the severity of any risk at a glance from its coordinates.

The power of treating these dimensions as independent is best seen through examples. Suppose the incident of concern is the **loss of data**. One cause might be a **fire** destroying the data centre: with proper security protections in place the probability of this is quite low, yet should it occur it could destroy the centre completely, giving a very high impact. The result is a **critical risk even though the probability is low**. The same loss of data might instead arise from **ransomware**: here the probability is higher than that of fire, while the impact depends on the type of data affected and may well be smaller. This gives a risk that is again medium-to-high but with a smaller impact. Because the two scenarios occupy different regions of the likelihood–impact space, the appropriate **countermeasures differ** — the organization may choose to act one way against fire and an entirely different way against ransomware. Different models exist that decompose probability into more granular dominating factors, but in general the two governing quantities remain likelihood and impact.

How to Reduce Risk

The aim of the whole governance loop is to **reduce the level of risk**, and since risk is the product of likelihood and impact, there are exactly three levers available. One may **reduce the potential impact** of the risk; one may **reduce the probability or frequency** of the risk; or one may apply **a combination of both**. Which lever is pulled, and how hard, is itself a cost–benefit decision.



Risk Management, ISO 27002, and Governance

ISO 27002 treats IT risk analysis as an **essential part of best practice in IT**. It holds that an organization must identify its security requirements, and it names three main sources of such requirements. The first is the **assessment of risks** to the organization, taking into account its overall business strategy and objectives; through a risk assessment, threats to assets are identified, vulnerability and likelihood of occurrence are evaluated, and potential impact is estimated. The second is the body of **legal, statutory, regulatory, and contractual requirements** that the organization, its trading partners, contractors, and service providers must satisfy, together with their socio-cultural environment. The third is the organization's own set of **principles, objectives, and business requirements** for handling, processing, storing, communicating, and archiving information in support of its operations. The standard stresses that the resources spent implementing controls must be balanced against the business harm that would likely result in their absence — once again the cost-effectiveness balance — and it points to ISO/IEC 27005 for detailed guidance on risk assessment, treatment, acceptance, communication, monitoring, and review.

A structural principle closes the topic. All risks that could possibly have a negative effect on the well-being of the organization, once they materialize, are definitively the **responsibility of management**. It follows that **all levels of**

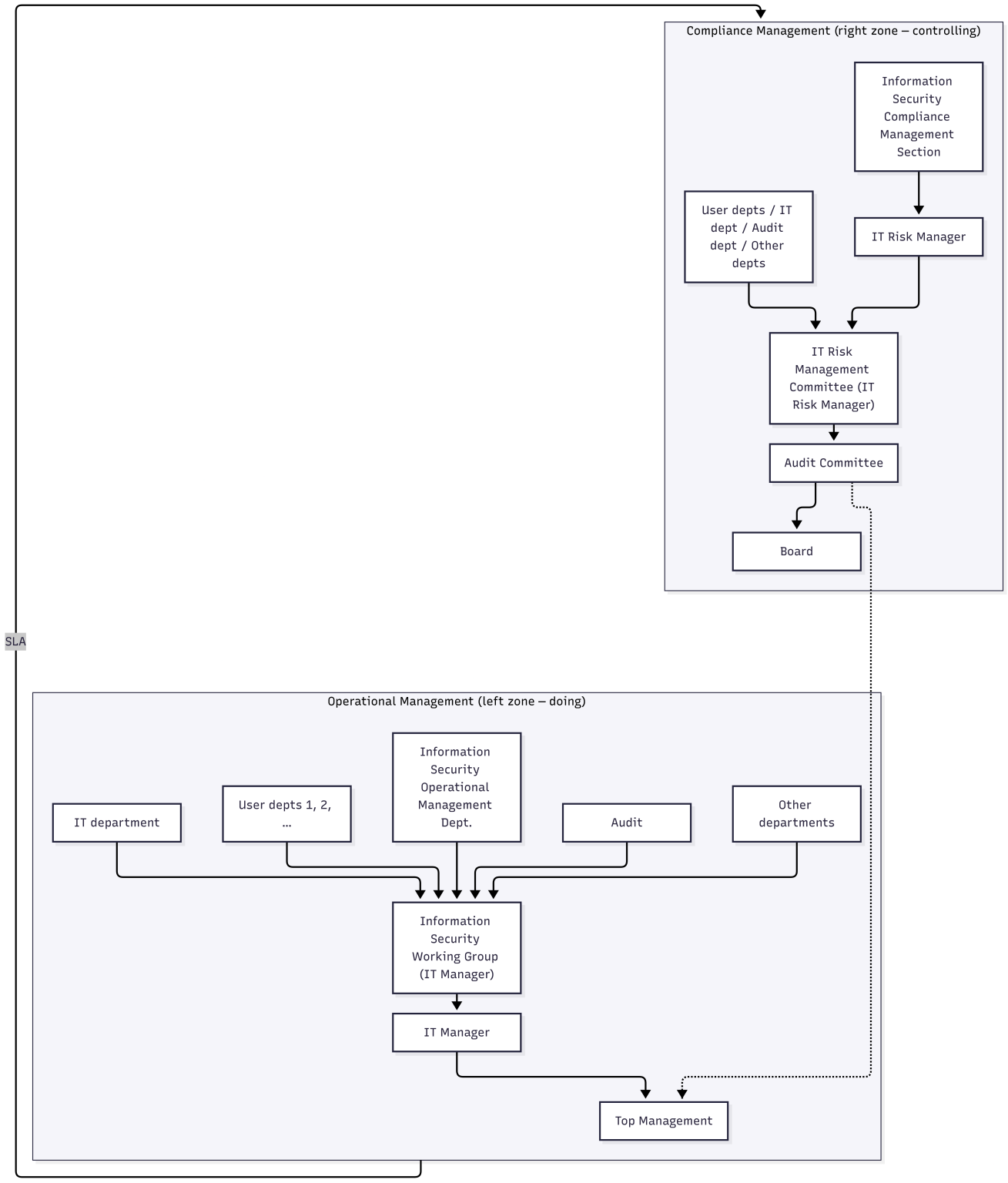
organizational management must be involved in the process of risk management, and that it is imperative for information and IT-related risks to be managed in an **integrated manner together with business risks** rather than in a technical silo. This is especially important in cybersecurity, because the people at the strategic level typically have no technical background; it therefore falls to the middle level to assess everything worth investigating, to highlight the critical issues, and to bring them to the attention of the managers who take decisions, since those managers must be made aware of all the potential bad situations that could occur.

Organization of Information Security

Two Orthogonal Security Functions

The way information security is organized inside a company is itself a matter of great importance, and every best practice document underlines this — ISO 27002, for instance, dedicates a clause titled *Organization of Information Security* to it. The guiding recommendation is that the organization should contain **at least two distinct components**, and that these two components are **orthogonal** to one another. The first looks after **day-to-day operational aspects** — operational security. The second is responsible for the **compliance monitoring function** — compliance management. The reason for the separation is structural: the people who *do* things must not be the same people who *check how things are done*, otherwise control would be grading its own work.

This produces a two-zone organizational chart. The **left zone**, Operational Management, is where the operational activities are actually carried out, implementing the bottom of the pyramid. The **right zone**, Compliance Management, is where the control process is implemented and feedback is provided back to the Direct process. The two zones meet at the top, at Top Management, and are connected to each other through a **Service Level Agreement (SLA)** that governs the relationship between what is operated and what is audited.



In the operational chart, an **Information Security Working Group** led by the IT Manager aggregates input from the user departments, the dedicated Information Security Operational Management department, the audit function, the IT department, and other departments, and reports upward through the IT Manager to Top Management. In the compliance chart, an **Information Security Compliance Management Section** feeds the **IT Risk Manager**, who sits on an **IT Risk Management Committee**; that committee collects input from all the departments, assembles a consolidated view — the matrix of how things are done — and passes it to an **Audit Committee** that reports to the Board. In

narrative terms, IT management on the operational side gathers information from the operational layer and composes it into a matrix view, which it hands to the person performing the audit; the audit side then collects input from across the departments and relays it to the risk management team, which in turn reports to top management.

Information Security Operational Management

The operational dimension is the set of people charged with ensuring that the system is functioning, organized into groups that monitor different departments, aggregate their information, and report to IT management. The activities it encompasses are the concrete, everyday actions of running security. They include **logical access control management**, meaning the real actions of adding to, changing, and deleting user access rights from access control lists. They include **identification and authentication management**, meaning the real actions of adding to, changing, and deleting entries in the user ID database and password files. They include **firewall management** — configuring firewalls with the authorized access rights, connecting workstations to LANs and to the Internet. They include **virus and malicious software management**, such as installing and updating antivirus software, together with the **handling of antivirus and related security incidents**. They include the **setting and updating of the security settings and configurations** of workstations and servers, the **ensuring of availability through UPS systems**, and the **ensuring of backups and the secure storage of those backups**, among other such day-to-day duties.

Information Security Compliance Management

The compliance dimension manages a different and orthogonal set of concerns, all directed at verifying that things are done in the right way and at assessing whether risk exists or persists. The activities that must be managed here include the **level to which previously identified IT risks are managed and mitigated**; the **level of information security awareness of users**; the **availability, completeness, and comprehensiveness of the information security policies, procedures, and standards**; the **level of compliance to those policies, procedures, and standards**; the **impact on the company's IT risk level when policies are not complied with**; the **compliance with regulatory, legal, and statutory requirements**; and **software licensing issues**, among others. In practice this side also takes the compliance clauses defined in the documents and attempts to apply them, verifying not only that

risks are controlled but that the whole documentary edifice is current, complete, and actually followed.

Awareness and the SETA Programme

Why Awareness Matters at Every Level

A recurring principle of the whole model is that **all levels are involved in the ISG process**, and awareness is the mechanism by which that involvement reaches the human core of the company. Information security procedures, guidelines, and practices must be drafted and conveyed to **all** users of information and IT in the organization, which means that every information worker must be made aware of — and trained in — the information security policy together with its associated procedures, guidelines, and practices. This is precisely why the board directive and CISP documents must be distributed as widely as possible: awareness is the capability of the company to carry a genuine security attitude, to recognize security issues, and to know how to handle them, and it cannot exist unless the knowledge is spread.

The SETA Programme and Its Three Temporal Horizons

To build this capability the model recommends developing a **Security Education, Training and Awareness (SETA)** programme — an instrument operating at the human level of the company, whose purpose is to inject the security factor into people. SETA should be an *extension* of the general knowledge employees already possess to do their jobs; its objective is to teach each of them to do that same job **securely**. The ambition is to move the awareness level from the basics to something that becomes automatic in the person's everyday life, creating a common understanding of both the issues and the solutions.

The programme has three stated objectives: to improve awareness of the importance and the need to protect organizational information resources; to acquire the necessary skills and know-how to do jobs more securely; and to create an understanding and insight into *why* it is important to protect organizational information assets. These three objectives map onto three distinct activities, each operating over a different time horizon, as captured by the differentiation between awareness, training, and education.

Awareness answers the attribute *what*, operates at the level of *information*, has the objective of making people *alert to* danger, is delivered through media such as videos, newsletters, and posters, and has a **short-term** impact — this is the first stage, developing the capability to understand what is dangerous. **Training** answers the attribute *how*, operates at the level of *knowledge*, has the objective of building *skill*, is delivered through practical instruction such as lectures, workshops, and hands-on practice, and has a **medium-term** impact — this is the stage that explains how to behave. **Education** answers the attribute *why*, operates at the level of *insight*, has the objective of *understanding*, is delivered through theoretical instruction such as seminars and literature study, and has a **long-term** impact — this is the most challenging stage, making people understand *why* they should behave in a certain way, and it is effective only over the long run.

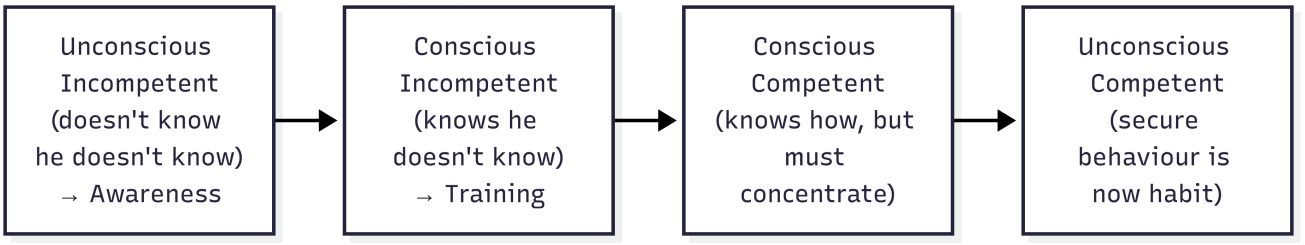
Attribute	Awareness	Training	Education
Attribute (question)	What	How	Why
Level	Information	Knowledge	Insight
Objective	Alert to	Skill	Understanding
Teaching method	Media (videos, newsletters, posters)	Practical instruction (lecture, workshop, hands-on)	Theoretical instruction (seminar, literature study)
Impact timeframe	Short-term	Medium-term	Long-term

The Conscious Competence Learning Model

The journey an employee takes through a SETA programme is described by the **Conscious Competence Learning Model**, a four-stage progression from ignorance to instinct.

The starting point is **Unconscious Incompetence**: the person does not even realize that they do not know how to do a certain thing safely. The priority here is to make the individual aware that they are incompetent as far as that specific task is concerned, and this is the province of **awareness** exercises. The second stage is **Conscious Incompetence**: the employee now knows that they lack the skill, and the priority becomes **training** them to do the job in a secure manner. The third stage is **Conscious Competence**: the employee knows what to do and how to do

it to keep the job secure, but still has to *concentrate* in order to perform the necessary procedures correctly. The fourth and final stage is **Unconscious Competence**: through enough practice and experience, all the security-related practices and skills have become inherent to the person's normal daily actions and behaviour — security has become a **habit** requiring no conscious effort.



The Pillars of Training

Underpinning the programme are the substantive topics that training must actually cover. At the level of *motivation*, employees must be taught why information is such an important asset, what the information security policy and procedures are, what the role and responsibility of every employee is, and what the consequences of not complying with the policy will be. At the level of *concrete practice*, the training should address user identification and authentication through passwords, how to choose a password, and the prohibition on sharing user IDs and passwords; virus control; backing up information; the sharing and storing of confidential information; secure email usage; secure Internet usage; social engineering; the handling of mobile devices such as notebooks and PDAs; the legal usage of software; office manners and discipline such as a clean-desk policy; and the required actions whenever a security incident or breach is suspected.

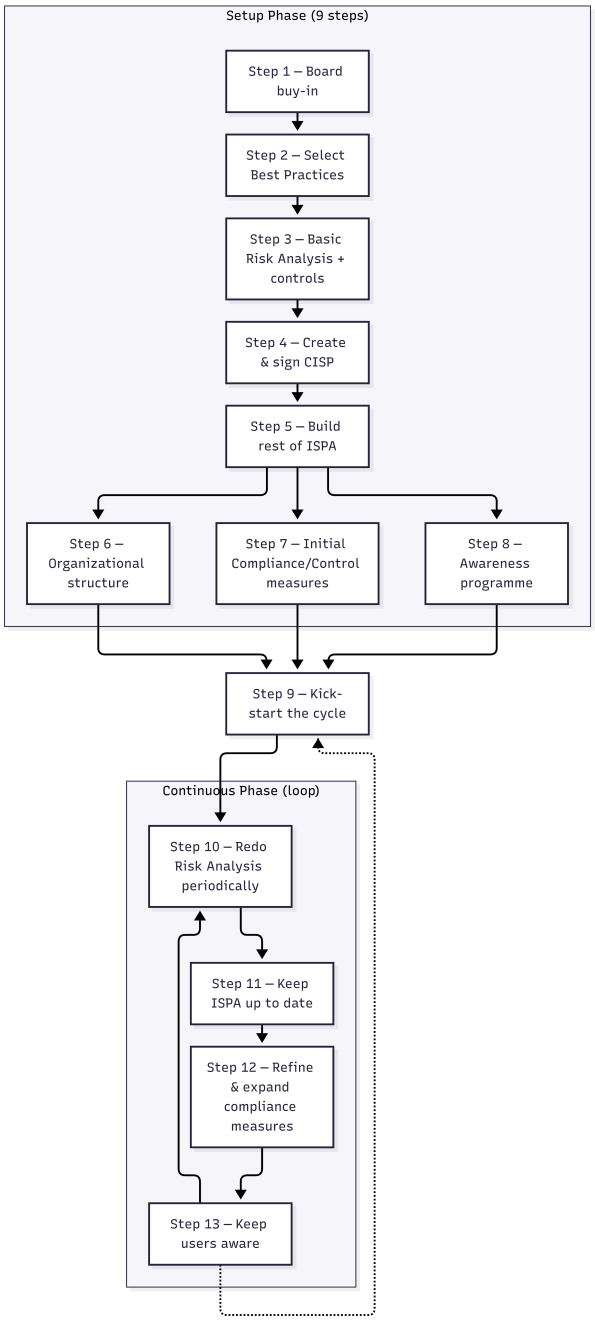
A Methodology for Establishing an ISG Environment

Putting the Whole Model Together

Having examined the Front Dimension, all five layers of the Depth Dimension, and the Best Practices basement, the final question is how to assemble the pieces into a working environment — *how do we put it all together?* The answer is a high-level **methodology of 14 steps**, deliberately written for the hardest case: it is a *complete* methodology that assumes no ISG environment exists at all and that everything must be started from scratch. In most real companies the situation is considerably better, so the methodology also carries a standing instruction to **re-use as much as possible** to avoid wasting time and money: the actual situation

must be compared against the steps, and must be synchronized and integrated with them wherever possible.

The 14 steps divide into two phases. The **Setup Phase** comprises the first nine steps (Steps 1 through 9), executed largely in sequence although Steps 6, 7, and 8 branch out in parallel from Step 5. The **Continuous Phase** comprises the final five steps (Steps 9 through 13, with Step 9 acting as the bridge that kick-starts the loop), which then repeat perpetually, implementing the Direct–Control flow on an ongoing basis.



The Setup Phase (Steps 1–9)

Step 1 — Get the Board's buy-in about IT risk management and information protection. This is a critically important starting point, and without it the project

will certainly struggle to progress properly. It is acceptable that one cannot initially obtain a *clean* Board Directive; what is essential is a genuine commitment from the board to support the project, which is enough to begin. Indeed, a basic, poor risk assessment may be performed first to identify the assets, under the approval of management, because the Direct process cannot start until it is known what must be protected. The formal Board Directive can come later — but it must be secured.

Step 2 — Select some guiding Best Practices. It is fundamental to choose an existing international best practice to serve as the foundation of the project and to provide motivation and direction. Many exist, and the essential point is simply to *have one*.

Step 3 — Perform a basic Risk Analysis and determine all needed controls. Some form of risk analysis is mandatory in order to identify the genuinely critical assets and to avoid wasting money and investment. This analysis requires involving a wide spectrum of people — board members, line management, users, system owners — and it must ensure that the relevant controls are installed and operational.

Step 4 — Create a CISP and have it signed by the Chairman/CEO. A CISP must be drafted, circulated among the stakeholders, and then submitted to top management. Recall that a more formal Board Directive on information security is essential in order to write a consistent CISP. The draft, once signed, becomes official company policy, and it forms the basis and motivation for all future steps.

Step 5 — Create the rest of the ISPA. The documents composing the full architecture must now be defined, taking as input the risk analysis from Step 3 and the CISP from Step 4, and following the guidelines discussed earlier. Throughout, the **connections between the documents must be preserved**, so that every sub-policy and procedure traces back up the tree.

Step 6 — Create the organizational structure for ISG. Specific attention must be paid to both the Operational Management and the Compliance Management sides, identifying the teams to be involved and the people responsible for them, and assigning responsibilities. Because this step is closely related to Step 7, it is preferable to perform the two **in parallel**.

Step 7 — Create an initial set of Compliance/Control measures and begin using them to produce reports on all three management levels. It is

important here to obtain buy-in from the company's internal and external IT audit departments as well as the legal department; if these were already involved in Step 3, the process is much easier. Creating such measures is not straightforward, and their effectiveness and value will have to be refined over time — the point is to start with an initial set that forms the basis of the Control part of the model.

Step 8 — Create an Awareness Programme, including aspects such as information security job responsibilities. This step is core to the success of the whole effort and must be performed on a continuous basis, with all the documents of the ISPA forming part of the programme.

Step 9 — Get the cycle going and kick-start the process. At this point the whole programme must be initiated; if one already exists, the revised programme must be integrated with the existing one to make the new one operational. Step 9 is the hinge between the two phases.

The Continuous Phase (Steps 10–13)

Step 10 — Redo the Risk Analysis from time to time to identify changes in risks and controls. Risks are dynamic: old ones disappear as patches are installed, ports are closed, and access controls are deployed; new ones materialize as new software is installed or configuration parameters change; and existing ones shift in impact, since a single mitigation action may decrease the impact of several other threats at once. Periodic reassessment keeps the risk picture current and the controls operational.

Step 11 — Keep the ISPA up to date and aligned with the newly identified risks, ensuring that all changed risks are reflected by updating the content of the ISPA and its compliance clauses where necessary.

Step 12 — Refine and expand the Compliance/Control measures to cater for newly identified risks, enforce compliance, and keep reporting to top management. The choice and creation of compliance clauses and monitoring measures remain challenging, and the process never stops: experience is gathered continuously and used to refine existing measures and to create new ones.

Step 13 — Continue to make all users more information-security aware. This process can never stop and must be enforced on a continuous basis, after which the loop returns to Step 10.

CHAPTER 2: Frameworks, Standards, Best Practices

1. The Problem of Governing Known Risk

In May 2017, hospitals across the United Kingdom discovered their computers displaying a single, chilling message: their files had been encrypted and would remain inaccessible unless a ransom was paid. The consequences were immediate and tangible. Patients were redirected to other facilities, appointments were cancelled, and several medical services were disrupted. The malware responsible was **WannaCry**, a ransomware worm that propagated rapidly across unpatched Windows machines.

The most instructive feature of this incident is not the sophistication of the attack, but its avoidability. The underlying Windows vulnerability had already been publicly disclosed, and Microsoft had released the corresponding security patch (MS17-010) *before* the attack took place. The vulnerability, in other words, was **known**. This raises the central question that motivates the entire discipline of security governance: if the risk was known and a remedy existed, why was the organization not prepared to manage it?

The answer is rarely technical. Patching a vulnerability is a solved problem in engineering terms. The failure lies instead in the absence of a **structured, organization-wide process** for discovering assets, assessing their exposure, prioritizing remediation, and ensuring that the remedy is actually applied across every machine. Managing known risk is a governance problem, and governance problems are addressed through frameworks, standards, and best practices. These three instruments form the conceptual backbone of this chapter.

2. Foundational Definitions

Before examining individual instruments, it is essential to distinguish three terms that are frequently used interchangeably in casual speech but carry precise and distinct meanings in security governance.

A **cybersecurity framework** is a structured, risk-based approach that helps an organization organize, assess, and improve its cybersecurity activities and outcomes. Its defining characteristics are that it is *structured* and *risk-based*. Being structured matters because it provides a **common language**: when multiple organizations describe their security posture using the same vocabulary and the

same categories, they can communicate, compare, and align with one another, which is indispensable for coordination and, ultimately, for certification. Being risk-based matters because it directs the organization toward doing the right things *before* damage occurs, rather than reacting after the fact. A framework provides guidance; it tells an organization what it should account for and how to think about security, without dictating the precise technical implementation.

A **cybersecurity standard** is a formal set of requirements or specifications that an organization can implement and audit against to demonstrate and certify conformity. The crucial distinction from a framework is formality and auditability. A standard contains specific requirements, and because those requirements are precise, an independent party can verify compliance and issue a certification. The canonical example is **ISO/IEC 27001**. Standards function as the foundation for governing security: they are standardized, so they supply a common language across organizations, and that shared language is precisely what makes certification meaningful.

A **best practice** is a procedure or method proven by research and experience to produce effective or optimal results and suitable for widespread adoption. Best practices are the most concrete of the three: they are *actionable* and *less formal* than standards. They describe proven actions an organization can take to implement what a framework recommends. Typical examples include patch management, multi-factor authentication, vulnerability disclosure, access-control hygiene, and user awareness training. Over time, best practices that prove their value tend to become embedded in standards and referenced by frameworks, so the three categories are not isolated but continuously feed into one another.

The relationship can be stated compactly: a **framework** provides guidance for managing cybersecurity risk, a **standard** provides specific requirements that can be implemented and assessed, and a **best practice** provides proven actions to implement. To illustrate the formality gradient with a mundane analogy, consider the "policy" of how to contact a professor. A policy is a high-level set of rules defining what is allowed and expected. The first rule might define the acceptable channel of contact (email); the second might require that the email be meaningful, with a clear leading subject; a third might require that the message provide sufficient context. This simple structure—high-level rules refined into specific, verifiable expectations—mirrors exactly how governance instruments descend from abstract guidance to concrete, checkable requirements.

3. Why Frameworks and Standards Matter

Frameworks and standards together provide a **roadmap to assess, improve, and communicate an organization's security posture**. They are, in the most fundamental sense, the foundations upon which security is governed. Three key aspects explain their enduring value. First, they offer **standardized approaches**, ensuring that different teams, departments, and even different companies operate from a shared understanding. Second, they are **risk-based and adaptable**, so that effort is directed where exposure is greatest and so that the same framework can serve organizations of radically different size and sector. Third, they institutionalize **continuous improvement**, treating security not as a project with an endpoint but as a process that is perpetually refined.

The most widely recognized instruments of this kind include the NIST Cybersecurity Framework (CSF), ISO/IEC 27001 and 27002, the CIS Controls, and COBIT, among many others.

4. Historical Evolution: From Standards to Frameworks to Best Practices

The modern landscape did not emerge all at once; it is the product of roughly three decades of accumulation, in which each layer built upon the previous one.

The lineage begins in the United Kingdom in the 1990s with **BS 7799**, a British information security standard that introduced one of the first structured approaches to information security management. In 1995, **BS 7799-1** was published as a code of practice—a catalogue of information security controls and guidance. In 1998, **BS 7799-2** added the management-system component, introducing the pivotal idea of a structured **Information Security Management System (ISMS)**: the notion that security should be managed through a deliberate, documented, continuously governed system rather than a set of ad hoc technical measures. In 2000, **ISO/IEC 17799** adopted this earlier British work at the international level, and in 2005 **ISO/IEC 27001** became the formal requirements standard for ISMSs, while **ISO/IEC 27002** evolved as its companion guidance on controls.

In parallel, best practices emerged as proven procedures that organizations could apply even before adopting a formal certification or framework. The authority that would come to formalize much of this thinking in the United States is the **National Institute of Standards and Technology (NIST)**, a federal agency founded in 1901 that develops scientific, technological, and measurement standards to promote innovation and industrial safety.

The framework era, as distinct from the standards era, crystallized in the mid-2010s. In 2013, Executive Order 13636 directed NIST to work with stakeholders to develop a *voluntary* framework for reducing cyber risk to critical infrastructure. In 2014, NIST issued Version 1.0 of the **Framework for Improving Critical Infrastructure Cybersecurity**, developed with extensive private-sector input and designed explicitly as voluntary guidance built on existing standards, guidelines, and practices—a design choice that made it flexible and broadly adoptable well beyond critical infrastructure. Also in 2014, the **Cybersecurity Enhancement Act** reinforced NIST's role, mandating support for the development of a voluntary, consensus-based, industry-led set of standards, guidelines, best practices, methodologies, procedures, and processes. This Act is historically significant because it explicitly connected the three concepts that remain central today: standards, frameworks, and best practices. In 2018, NIST released **CSF Version 1.1**, adding cyber supply-chain risk management, self-assessment guidance, and clarifications on identity-related concepts. The NIST CSF is today recognized worldwide as a basis of security governance.

5. The Organizational Landscape and the Two Core Layers

The ecosystem of frameworks is vast. Beyond the general-purpose instruments, there exist sector-specific and purpose-specific frameworks: DORA and the SWIFT Customer Security Programme for finance, NIS2 and the EU Cybersecurity Act for the European Union, FISMA and FedRAMP for U.S. federal and cloud systems, CMMC for the U.S. defense supply chain, the HIPAA Security Rule for healthcare, PCI DSS for payment cards, IEC 62443 for industrial and operational technology, TISAX for automotive, and NERC CIP for the electricity sector. National frameworks exist as well, including the Italian National Cybersecurity Framework, Germany's BSI IT-Grundschutz, France's ANSSI guidance, the UK's NCSC Cyber Assessment Framework, and Australia's Essential Eight. Further specialized families cover application security (OWASP SAMM, OWASP Top 10, BSIMM, NIST SSDF), cloud security (CSA Cloud Controls Matrix, ISO/IEC 27017, FedRAMP), Zero Trust (NIST SP 800-207, CISA Zero Trust Maturity Model), maturity models (C2M2, CMMC, COBIT), and threat-modeling and intelligence schemes (MITRE ATT&CK, the Cyber Kill Chain, STRIDE, STIX/TAXII). The sheer breadth of this catalogue is itself a lesson: there is no single instrument, and competent governance means selecting and combining the right ones.

For the purposes of structured study, the core instruments can be organized into two complementary layers that answer two different questions.

The **framework layer** answers the strategic question: *what should I account for, and how should I organize security?* It provides principles and practices for managing and improving security. Its representatives are **ISO 27001, the NIST CSF, and the Italian Cybersecurity Framework**.

The **implementation layer** answers the operational question: *how do I apply the strategy?* It provides specific safeguards and measures to mitigate security risks. Its representatives are **ISO 27002, NIST SP 800-53, and the CIS Controls**.

Together, these two layers constitute **security governance**. A finer-grained mapping sharpens the distinction between the instruments and the questions each one is best suited to answer:

Layer	Main question	Instruments
Management system	How should information security be managed continuously?	ISO/IEC 27001
Control guidance	Which information security controls can be implemented?	ISO/IEC 27002
Risk framework	How can cybersecurity risk be organized and communicated?	NIST CSF, Italian NCF
Detailed controls	Which safeguards can be selected, tailored, and assessed?	NIST SP 800-53
Practical safeguards	What should be prioritized and implemented first?	CIS Controls

The logical progression of this chapter follows these layers. First come the frameworks—principles and standards giving the general idea of what to do within an organization. Then comes implementation—how to satisfy the security requirements the framework demands. Once implementation is understood, the governance pyramid can be reassembled in full.

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6. ISO/IEC 27001

ISO/IEC 27001 is the first and best-known standard in the ISO 27000 family, and it is the world's foremost standard for Information Security Management Systems. It is the most international of the governance instruments, recognized in over sixty countries, which is precisely what gives it its value as a lingua franca of security certification.

6.1 The ISO 27000 Family

ISO 27001 does not stand alone; it sits at the head of a large family of related standards, each addressing a facet of information security management. ISO/IEC 27000 provides the overview and vocabulary; **27001** specifies the requirements for an ISMS; **27002** supplies the code of practice for implementing the controls; 27003 gives implementation guidance; 27004 addresses measurement; 27005 addresses information security risk management; and 27006 and 27007 deal with audit and certification bodies. Further members extend the family into specialized domains—27011 for telecommunications, 27017 and 27018 for cloud and personally identifiable information in the cloud, 27031 for ICT business-continuity readiness, 27032 for cybersecurity, the 27033 series for network security, 27035 for incident management, 27037 for digital evidence, and 27799 for health information. For the study of governance, the two members that matter most are **27001**, which defines *what* must be managed, and **27002**, which explains *how* the controls can be implemented.

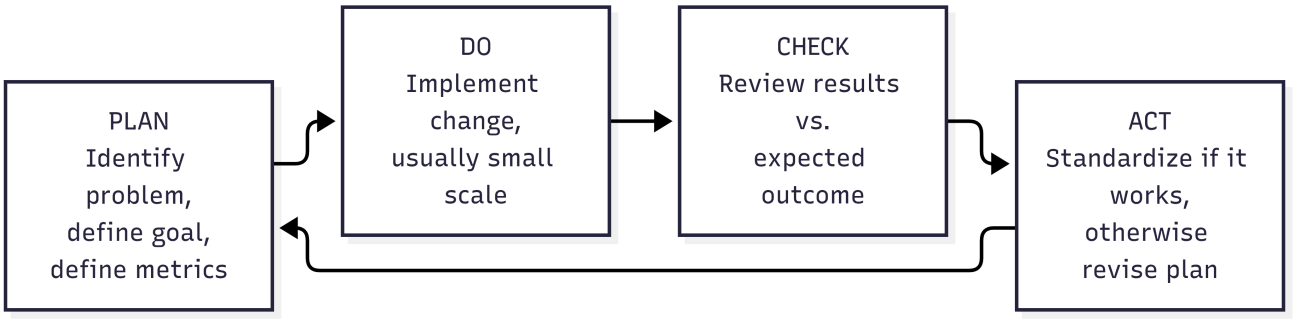
6.2 The Management-System Approach

The single most important conceptual property of ISO 27001 is that it follows a **management-system approach**. This means security is treated as an **ongoing organizational process** rather than a one-time technical project. The standard specifies the requirements for establishing, implementing, maintaining, and *continually improving* an ISMS. Its purpose is to ensure that information security is managed systematically through governance, risk assessment, controls, monitoring, and improvement. Concretely, an organization adopting ISO 27001 must define its scope, assign responsibilities, assess and treat its risks, measure its performance, and continually improve the system. The requirements it specifies are known as **security controls**, and organizations are expected to be compliant with them. These controls are deliberately not highly detailed, because they must cover a very wide range of topics: the scope and objectives of the organization, the human dimension, the assessment and management of risk, the measurement of performance, and the continuous improvement of the ISMS.

6.3 The Deming Cycle (PDCA)

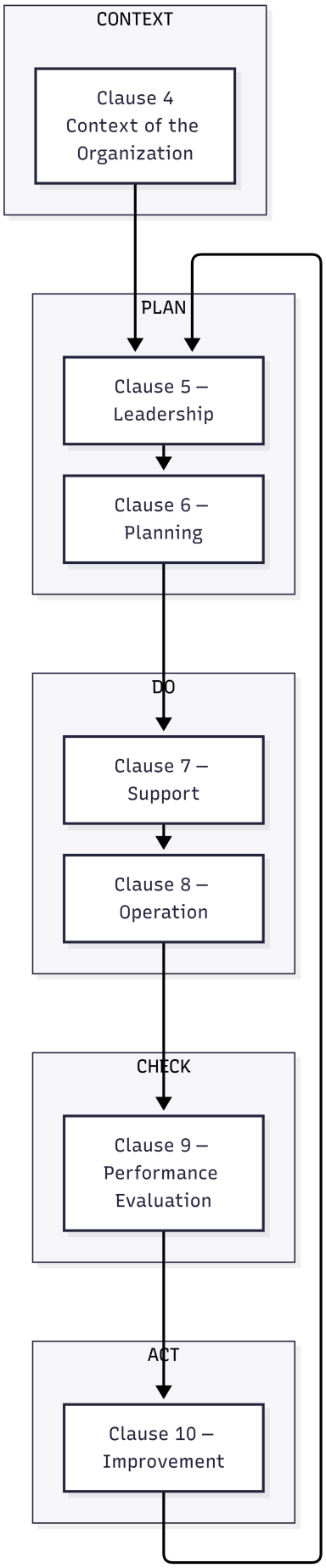
The engine that drives the "continual improvement" character of ISO 27001 is the **Deming Cycle**, also known as **PDCA**, a continuous-improvement method used to test changes in a structured way. Its four phases form a loop that never terminates.

In the **Plan** phase, the organization identifies a problem, defines a goal, and decides in advance how success will be measured. In the **Do** phase, it implements the change, typically on a small scale to limit exposure. In the **Check** phase, it reviews the results and compares them against the expected outcome, often on the basis of concrete measurements. In the **Act** phase, it standardizes the improvement if it worked, or revises the plan and repeats the cycle if it did not. The iterative nature of the loop is the entire point: security is never "finished," and each pass through the cycle refines the system a little further.



6.4 The Structure of ISO 27001: Clauses

ISO 27001 is organized into **clauses**, each of which is a section of the document. Clauses 1 to 3 are introductory, providing context and definitions. The **mandatory requirements reside in Clauses 4 to 10**, and these seven clauses map directly onto the phases of the Deming Cycle. Clause 4, the context of the organization, underlies and informs every phase of the cycle. Clauses 5 and 6 (Leadership and Planning) constitute the *Plan* phase. Clauses 7 and 8 (Support and Operation) constitute the *Do* phase. Clause 9 (Performance Evaluation) is the *Check* phase. Clause 10 (Improvement) is the *Act* phase. The standard is thus organized *across* the cycle: context covers all the steps, and the remaining clauses occupy their positions around the loop.



Each clause deserves examination in its own right, because together they define the complete anatomy of a managed security system.

Clause 4 — Context of the Organization. The organization must understand itself, the interested parties around it, and the scope of its ISMS. This clause establishes the boundaries within which everything else operates: what is being protected, who cares about it, and how far the management system reaches.

Clause 5 — Leadership. This clause makes information security a *leadership responsibility, not merely a technical activity.* It contains three sub-requirements. Under *5.1 Leadership and commitment*, top management must actively support the ISMS, align it with the business strategy, provide the necessary resources, and ensure the system achieves its intended outcomes. Under *5.2 Policy*, the organization must define and communicate an information security policy that sets direction, supports objectives, and reflects a commitment to compliance and continual improvement. Under *5.3 Organizational roles, responsibilities, and authorities*, roles must be assigned clearly so that responsibilities are understood and reporting lines are effective. A recurring and non-negotiable principle of the standard appears here: **every person must be assigned a role and a responsibility, and this assignment must be formalized.**

Clause 6 — Planning. This clause turns context and leadership into a practical plan for managing risk and improving security. Under *6.1 Actions to address risks and opportunities*, the organization identifies the risks and opportunities related to the ISMS and plans actions to reduce threats, prevent problems, and improve effectiveness—this is where the security controls needed to assess and secure risks are brought to bear. Under *6.2 Information security objectives and planning to achieve them*, the organization sets **measurable** security objectives and defines who will do what, by when, and with which resources. Under *6.3 Planning of changes*, changes to the ISMS are managed in a controlled way so that security is not weakened by poorly planned updates. Such change management may be performed **manually**, or **automatically** through self-adaptive systems that detect and respond to vulnerabilities on their own.

Clause 7 — Support. This clause ensures the ISMS is supported by competent people, clear communication, and proper documentation. It concerns resources, competence, awareness, communication, and documented information. Under *7.1 Resources*, the organization provides the people, tools, budget, and infrastructure needed to run the ISMS. Under *7.2 Competence*, it ensures that those performing security-related work have the necessary knowledge, training, and experience. Under *7.3 Awareness*, it makes sure staff understand the security policy, their responsibilities, and the consequences of non-compliance. Under *7.4 Communication*, it defines what must be communicated, to whom, when, and

through which channels—exactly the kind of formalized communication exemplified earlier by the rules governing a meaningful, context-rich email. Under *7.5 Documented information*, it creates, maintains, and controls the documents and records needed to support and to *prove* that the ISMS works.

Clause 8 — Operation. This is the clause where the ISMS becomes operational—where risks are assessed, treatments are chosen, and controls are applied in practice. Under *8.1 Operational planning and control*, operational processes are planned and controlled so the ISMS is implemented as intended, with explicit reference to the implementation guidance of ISO 27002. Under *8.2 Information security risk assessment*, a consistent method for identifying and evaluating risks is defined. Under *8.3 Information security risk treatment*, controls or measures are selected and implemented to treat the risks deemed **unacceptable**.

Clause 9 — Performance Evaluation. This clause checks whether the ISMS is delivering results and whether leadership is using the evidence to steer improvements. Under *9.1 Monitoring, measurement, analysis, and evaluation*, the organization tracks performance using indicators, metrics, and analysis. Under *9.2 Internal audit*, it regularly checks whether the ISMS conforms both to ISO 27001 and to the organization's own requirements, continuously auditing and searching for evidence of risk. Under *9.3 Management review*, top management reviews the ISMS to confirm it remains suitable, effective, and aligned with business needs. Notably, even here the standard remains non-prescriptive about exact technical steps: it tells the organization *that* it must monitor, audit, and review, without dictating the precise procedure.

Clause 10 — Improvement. This clause concerns conformity and the organization's capacity to improve continuously, including how it reacts after an incident. Under *10.1 Nonconformity and corrective action*, the organization investigates problems, fixes their root causes, and prevents recurrence. Under *10.2 Continual improvement*, it keeps strengthening the ISMS over time on the basis of audits, incidents, measurements, and changing risks.

6.5 Annex A and the Statement of Applicability

Beyond the ten clauses, ISO 27001 includes **Annex A**, a reference set of information security controls that functions as a **checklist**. In the 2022 revision, Annex A contains exactly **93 controls**. Organizations are not required to implement all of them indiscriminately; instead, they evaluate each control's applicability on the basis of their risk-treatment decisions. These decisions are

documented in a central artefact called the **Statement of Applicability (SoA)**, which links risks, selected controls, and the justification for each inclusion or exclusion.

A well-constructed SoA records, for each control, the Annex A control identifier and name, whether the control is applicable, a justification for its inclusion or exclusion, and the implementation status of the controls deemed applicable. It normally also references the relevant policies, procedures, or evidence. The SoA is thus the document that connects abstract risk decisions to concrete controls and to the proof that those controls exist.

The relationship between Annex A and ISO 27002 is the natural bridge to the implementation layer: **ISO 27001 (through Annex A) defines *what* the 93 controls are, while ISO 27002 defines *how* each of those controls can be implemented.** ISO 27002 begins precisely from the controls catalogued in Annex A and expands each one into detailed guidance.

6.6 Could ISO 27001 Have Prevented WannaCry?

Returning to the motivating incident with the apparatus of ISO 27001 in hand reveals a subtle but critical lesson about the limits of certification. Imagine being the CISO of an organization with 2,000 Windows computers at the moment WannaCry strikes, having already implemented ISO 27001, and discovering that hundreds of machines remain unpatched despite Microsoft's prior release of MS17-010. How could this happen?

The decisive insight is that **possessing an ISO 27001 certificate does not prove that every computer was patched.** A certificate attests that a *management system* exists and functions, not that every technical control is perfectly executed at every instant on every asset. The standard, properly applied, nonetheless supplies the activities that *should* have prevented the situation: maintaining an **asset inventory** so that every Windows system is known; performing **risk assessment** so that the exposure is recognized; operating **vulnerability management** so that the available patch is identified and applied; conducting **monitoring** so that unpatched machines are flagged; preparing **incident response** so that an outbreak can be contained; and sustaining **security awareness** so that people understand their obligations. The failure in a real organization is therefore not the absence of the framework but the gap between holding a certificate and genuinely operating every clause and control the

framework demands—most of all the Clause 8 operational discipline that turns risk treatment into applied, verified safeguards.

7. The NIST Cybersecurity Framework (CSF) 2.0

If ISO 27001 is the most *international* governance instrument, the **NIST Cybersecurity Framework** is the most *widely used* security governance framework in the world. Where ISO 27001 is a certifiable standard built around a management system, the NIST CSF is a voluntary, non-certifiable framework built around the organization and communication of risk. The two are complementary rather than competing, and much of the CSF's power comes from its flexibility and its universal applicability.

The framework focuses on using **business drivers** to guide cybersecurity activities and on treating cybersecurity risk as an integral part of the organization's broader risk-management processes. Although it was originally developed to improve cybersecurity risk management in critical infrastructures, it is deliberately constructed so that any organization, in any sector or community, can adopt it. It provides a common organizing structure that assembles standards, guidelines, and practices already shown to work effectively in the field. Its applicability spans every kind of technology-dependent organization, whether the focus is on information technology (IT), industrial control systems (ICS), cyber-physical systems (CPS), or the broader world of connected devices and the Internet of Things (IoT).

The framework consists of three parts, which together answer three different governance needs: the **Core**, the **Implementation Tiers**, and the **Profiles**.

7.1 Features of the NIST CSF

The defining feature of the CSF is that it is **technology-neutral**. It remains effective regardless of the specific technologies in use and actively supports technical innovation, because it references a variety of existing standards, guidelines, and practices that themselves evolve alongside technology. Rather than prescribing particular products or configurations, it describes outcomes.

This neutrality enables the framework to provide a **common taxonomy and mechanism** through which an organization can accomplish five distinct things. It can describe its *current* cybersecurity posture; it can describe its *target* state for cybersecurity; it can identify and prioritize opportunities for improvement within a continuous and repeatable process; it can assess its progress toward the target

state; and—recurring once more as a central theme of all governance—it can **communicate** about cybersecurity risk among internal and external stakeholders. Communication is not a peripheral benefit but one of the explicit purposes the framework is designed to serve.

7.2 The Core and Its Four Elements

The **Core** is the heart of the framework. It provides a set of activities to achieve specific cybersecurity outcomes and references examples of guidance for achieving those outcomes. A crucial clarification distinguishes the CSF Core from an ISO-style checklist: **the Core is not a checklist of actions to perform.** Instead, it presents key cybersecurity *outcomes*, identified by stakeholders as helpful in managing cybersecurity risk. The difference is more than semantic. A checklist enumerates tasks to tick off; an outcome catalogue describes states to achieve, leaving the organization free to determine how best to reach them.

The Core is built from four nested elements, descending from the most general to the most specific.

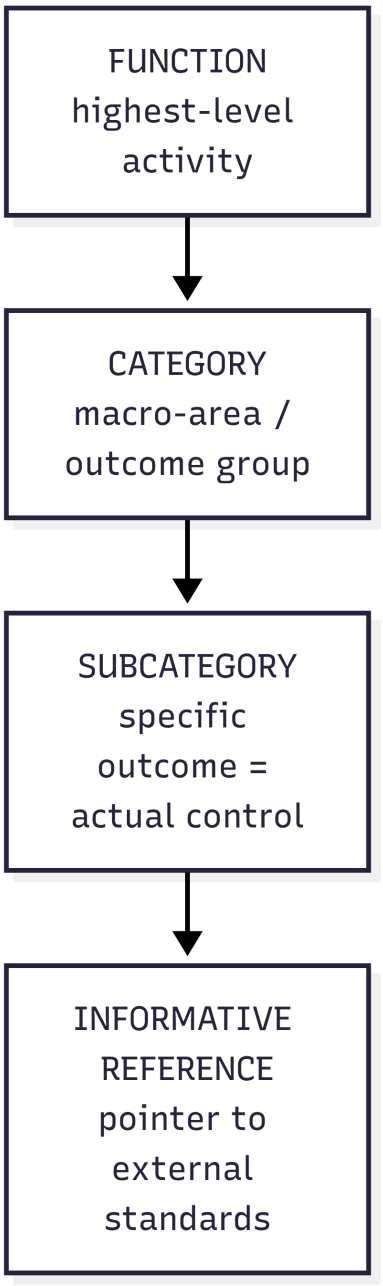
Functions organize basic cybersecurity activities at their highest level. They are the broad pillars of the framework.

Categories are subdivisions of a Function into groups of cybersecurity outcomes closely tied to programmatic needs and particular activities. They represent the **macro-areas** in which a Function operates. Examples include "Asset Management," "Identity Management and Access Control," and "Detection Processes."

Subcategories further divide a Category into specific outcomes of technical or management activities. They are the level at which the framework specifies the concrete actions to be taken—the actual security controls—and they provide the set of results that support the achievement of each Category's outcomes. Examples include "External information systems are catalogued," "Data-at-rest is protected," and "Notifications from detection systems are investigated."

Informative References are specific sections of standards, guidelines, and practices—common among critical-infrastructure sectors—that illustrate concrete methods for achieving the outcomes associated with each Subcategory. They are, in effect, **pointers to other standards** that complement and operationalize the

information contained in the Categories and Subcategories.



7.3 The Six Functions

In its original form the CSF comprised five Functions. The 2.0 revision introduced a sixth, **Govern**, which occupies a special position: whereas the other five Functions are each specific to particular operational aspects of cybersecurity, Govern is cross-cutting and *covers all the others*. Governing security means, in essence, possessing expertise across every function, with responsibility threading through the entire process—which is precisely why a standardized, structured approach is indispensable.

The six Functions, with the organizational understanding each one develops, are as follows. **Identify (ID)** develops the organizational understanding needed to manage cybersecurity risk to systems, people, assets, data, and capabilities—it

supplies the controls necessary for understanding and deciding an organization's starting objectives. **Protect (PR)** develops and implements the appropriate safeguards to ensure delivery of critical services. **Detect (DE)** develops and implements the activities needed to identify the occurrence of a cybersecurity event. **Respond (RS)** develops and implements the activities to take action regarding a detected incident. **Recover (RC)** develops and implements the activities to maintain resilience plans and restore any capabilities or services impaired by an incident. **Govern (GV)** establishes and monitors the organization's cybersecurity risk-management strategy, expectations, and policies, guiding, prioritizing, and overseeing all the other activities.

Each Function is examined below in terms of its Categories.

Govern (GV). Govern establishes the strategic conditions that make the other five functions effective. Its Categories are *Organizational Context (GV.OC)*, understanding the organization's mission, stakeholders, dependencies, and legal, regulatory, and contractual requirements; *Risk Management Strategy (GV.RM)*, establishing priorities, constraints, risk appetite, risk tolerance, and assumptions for cybersecurity decisions; *Roles, Responsibilities, and Authorities (GV.RR)*, defining accountability, decision-making authority, and responsibilities; *Policy (GV.PO)*, establishing, communicating, and enforcing cybersecurity policies; *Oversight (GV.OV)*, using risk-management results and performance information to review and adjust the strategy; and *Cybersecurity Supply Chain Risk Management (GV.SC)*, identifying, assessing, managing, monitoring, and improving the risks arising from suppliers and third parties.

Identify (ID). The governing maxim of this Function is that *you cannot protect what you have not identified.* Its Categories are *Asset Management (ID.AM)*, inventorying and managing data, devices, systems, and services; *Risk Assessment (ID.RA)*, identifying and evaluating cybersecurity risks; and *Improvement (ID.IM)*, using lessons learned to improve risk management.

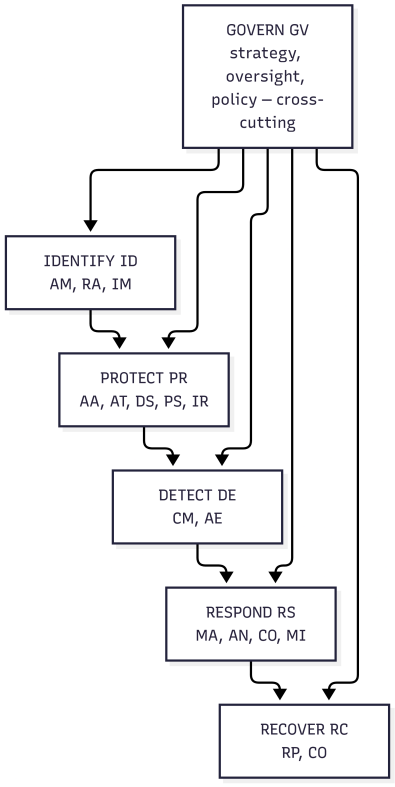
Protect (PR). The core principle here is to *reduce the likelihood and impact of an attack.* Its Categories are *Identity Management, Authentication, and Access Control (PR.AA)*, managing identities, authentication, and permissions; *Awareness and Training (PR.AT)*, training users and partners to perform their cybersecurity duties; *Data Security (PR.DS)*, protecting the confidentiality, integrity, and availability of data; *Platform Security (PR.PS)*, hardening and securing platforms and devices; and *Technology Infrastructure Resilience (PR.IR)*, ensuring the infrastructure can withstand and recover from disruption.

Detect (DE). The core principle is that *faster detection means faster response and less damage.* Its Categories are *Security Continuous Monitoring (DE.CM)*, continuously monitoring systems and networks, and *Adverse Event Analysis (DE.AE)*, identifying and analyzing abnormal activity.

Respond (RS). The governing idea is that *a good response reduces the operational and reputational impact of an incident.* Its Categories are *Incident Management (RS.MA)*, maintaining and executing incident-response procedures; *Incident Analysis (RS.AN)*, analyzing the incident to support response and recovery; *Incident Response Reporting and Communication (RS.CO)*, coordinating internal and external communications; and *Incident Mitigation (RS.MI)*, containing, eradicating, and reducing the incident's impact.

Recover (RC). The essential insight is that *recovery is not only restoration but also improvement.* Its Categories are *Incident Recovery Plan Execution (RC.RP)*, executing recovery processes and restoring services, and *Incident Recovery Communication (RC.CO)*, communicating recovery status and return-to-normal progress.

The complete Core can be visualized as the six Functions wrapped around a single wheel, with Govern at the center informing all the others:



7.4 Implementation Tiers

The second part of the framework, following the Core, is the set of **Implementation Tiers**. The Tiers provide context on *how an organization views cybersecurity risk* and on the processes it has in place to manage that risk. They range from **Tier 1 (Partial)** through **Tier 2 (Risk Informed)** and **Tier 3 (Repeatable)** to **Tier 4 (Adaptive)**, describing an increasing degree of rigor and sophistication in risk-management practices.

Two caveats are essential to interpreting the Tiers correctly. First, the Tiers give an indication of how an organization has implemented its practices and are useful for supporting improvement, but they **do not provide any certification**. Second, and emphatically, **the Tiers do not represent maturity levels.** An organization at Tier 1 is encouraged to consider moving toward Tier 2 or higher, but the purpose of the Tiers is to support decision-making about how to manage cybersecurity risk and about which dimensions of the organization are higher priority and deserve additional resources. Progression to a higher Tier is encouraged only when a cost-benefit analysis indicates that it would achieve a feasible and cost-effective reduction of cybersecurity risk—climbing the Tiers is a deliberate economic choice, not an automatic goal.

The character of each Tier can be understood across three dimensions: the **risk-management process**, the **integrated risk-management program**, and the degree of **external participation**.

At **Tier 1 — Partial**, the risk-management process lacks formalization and is managed on an ad hoc, sometimes reactive basis; the prioritization of activities may not be driven by organizational risk objectives, the threat environment, or business requirements. There is limited awareness of cybersecurity risk at the organizational level, risk is managed irregularly and case by case according to varied experience, and the organization may lack information-sharing processes altogether. External participation is characterized by **no collaboration**.

At **Tier 2 — Risk Informed**, risk-management practices are approved by management but may lack clearly articulated policies, and prioritization still may not flow directly from organizational risk objectives. There is awareness of cybersecurity risk at the organizational level, but no organization-wide approach to managing it; information is shared internally on an informal basis, not all levels of the organization are involved, and risk assessments of internal and external assets occur but are not typically repeatable or recurring. External participation is characterized by **low collaboration**.

At **Tier 3 — Repeatable**, policy is formally approved and defined, and it is regularly updated; there exists an organization-wide approach to managing cybersecurity risk. External participation rises to **high collaboration**.

At **Tier 4 — Adaptive**, standards and best practices are applied, and the organization runs a continuous-improvement process; its organization-wide approach uses risk-informed policies, processes, and procedures to address potential cybersecurity events as they evolve. External participation reaches **full collaboration**.

	Risk Management Process	Integrated Program	External Participation
Tier 1 — Partial	Ad hoc, reactive, not formalized	Limited organization-level awareness	None
Tier 2 — Risk Informed	Management-approved, policies may be absent	Aware but not organization-wide; informal sharing	Low
Tier 3 — Repeatable	Formally approved policy, regularly updated	Organization-wide approach	High
Tier 4 — Adaptive	Standards and best practices, continuous improvement	Organization-wide, risk-informed, event-adaptive	Full

7.5 Profiles

The third part of the framework is the **Profile**. A Framework Profile is the alignment of the Functions, Categories, and Subcategories with the particular business requirements, risk tolerance, and resources of an organization. Profiles can describe either the current state or the desired target state of specific cybersecurity activities. The **Current Profile** indicates the cybersecurity outcomes that are presently being achieved, while the **Target Profile** indicates the outcomes needed to meet the organization's desired risk-management goals. The comparison between the two is what enables a **gap analysis**: the difference between where the organization is and where it wishes to be defines the work that remains. Importantly, the framework does not prescribe any Profile templates, preserving flexibility in how the comparison is implemented.

7.6 How to Use the NIST CSF

The framework supports a range of practical uses within an organization. It can serve as a basic review of existing cybersecurity practices; as the foundation for establishing or improving a cybersecurity program; as a vehicle for communicating cybersecurity requirements with stakeholders; as an input to buying decisions; as a means of identifying opportunities for new or revised informative references; as a methodology for protecting privacy and civil liberties; and as an instrument for self-assessing cybersecurity risk.

7.7 The Framework in Action: A Worked Example

The abstract machinery of the CSF becomes concrete when applied to a scenario. Consider a **university research department** that wishes to protect sensitive research data, cloud services, staff accounts, and laboratory systems. The first governance step is to define the **organizational scope**, which in this case encompasses the research servers and laboratory systems; the cloud storage and collaboration platforms; the research data and personal information; the staff, researchers, students, suppliers, and external partners; and the legal, contractual, and university security requirements that apply.

With scope established, the department constructs its **Profiles** by examining each relevant Function and Category, recording the current state, the target state, and the priority. The result is a structured gap analysis of the kind shown below.

Function and category	Current state	Target state	Priority
GV.OC — Organizational Context	Critical services and external dependencies not fully documented	Document critical services, stakeholders, dependencies, requirements	High
GV.RR — Roles and Responsibilities	Security responsibilities informally assigned to IT staff	Approve clear responsibilities across management, IT, researchers, suppliers, IR	High
GV.SC — Supply Chain Risk Mgmt	Cloud and software suppliers reviewed inconsistently	Establish supplier security requirements, due diligence, monitoring, obligations	High

Function and category	Current state	Target state	Priority
ID.AM — Asset Management	Hardware inventoried; cloud services and datasets incomplete	Maintain updated inventory with owners, classifications, criticality	High
ID.RA — Risk Assessment	Performed only after major incidents or projects	Perform periodic and event-driven assessments via documented method	High
PR.AA — Access Control	MFA protects admins but not all users	Require MFA for all accounts; review privileged access regularly	High
PR.DS — Data Security	Backups exist; encryption and restore testing vary	Encrypt sensitive data, define retention, test restoration regularly	High
DE.CM — Continuous Monitoring	Logs collected from selected systems, reviewed manually	Centralize logs, generate alerts for suspicious activity	Medium
RS.MA — Incident Management	General IT procedures only; no research-specific playbook	Establish, test, maintain IR procedures and escalation paths	High
RC.RP — Recovery Plan Execution	Services restorable but priorities and objectives unclear	Define recovery priorities, responsibilities, RTOs, RPOs, exercises	Medium

The department then situates itself on the **Tier** scale. It is best characterized as **Tier 2 — Risk Informed**, because it recognizes cybersecurity risks and performs some risk-management activities, but its practices are not yet consistently established or applied across the organization. The evidence for this classification is telling: security risks are discussed in IT and project meetings; some systems use MFA, backups, logging, and access reviews; risk assessments occur for selected projects or incidents; supplier risks are known but not managed through a consistent process; cybersecurity information flows mainly through informal communication; and procedures vary between laboratories and research teams.

The chosen **target is Tier 3 — Repeatable**, in which risk-management practices are formally approved, documented, consistently implemented, and regularly reviewed across the department. Reaching Tier 3 requires a concrete set of actions: approving an organization-wide risk-management policy; assigning formal roles and responsibilities; maintaining complete asset and data inventories; applying a single risk-assessment method to all relevant systems; requiring MFA and periodic access reviews; standardizing supplier-security requirements; centralizing monitoring and incident reporting; testing incident-response and disaster-recovery plans; and using metrics and management reviews to track improvement.

7.8 What Tier Was WannaCry?

The framework also offers a diagnostic lens on the WannaCry failure. Reconsider the organization in the aftermath of the attack, having discovered that IT knew some machines were unpatched, that there was no complete inventory of Windows systems, that patching was performed manually whenever administrators happened to have time, that the security team had previously warned management about the risk, and that there was no organization-wide process for prioritizing vulnerabilities—prompting management to ask, "How did we not know this was going to happen?"

This profile corresponds unmistakably to **Tier 1 — Partial**. The reactive, ad hoc patching, the absence of a complete inventory, the lack of any organization-wide vulnerability-prioritization process, and the informal, unheeded internal warnings are the textbook signature of the Partial Tier. Improving beyond it would mean precisely the moves that define the ascent toward Tier 3: building a formal asset inventory, instituting a documented and repeatable vulnerability-management process, assigning clear responsibilities, and establishing an organization-wide, management-endorsed approach to prioritizing and treating risk.

8. The Italian National Cybersecurity Framework (NCF)

Having examined one international standard (ISO 27001) and one international framework (NIST CSF), the natural next step is a **national** framework: the Italian National Cybersecurity Framework, formally the *Framework Nazionale per la Cybersecurity e la Data Protection*. Its existence illustrates an important principle of governance—that a framework may need to be adapted to the

economic and institutional realities of the country in which it operates, without abandoning international compatibility.

8.1 Motivations

The Italian NCF was created to pursue several interlocking objectives, each responding to a specific feature of the national context.

The first and most elevated aim is to bring **cyber-risk awareness to the highest levels of the business**, so that cybersecurity is no longer treated merely as a matter for technical staff. The framework seeks to make organizations fold cyber risk into their general risk-management processes, treating it as a component of *economic* risk rather than a purely technical nuisance.

This ambition is shaped decisively by the **structure of the Italian economic landscape**. A striking **69% of the national GDP** is generated by small and medium-sized enterprises (SMEs), and there are very few large national enterprises —on the order of **0.1%**. A framework intended for Italy therefore cannot be designed around the resources and sophistication of large corporations; it must be usable by a vast population of small organizations with limited budgets and limited specialist staff.

The framework further aims to create something **recognized at the international level**, to **improve information-sharing capacity**, and to **increase the duty of care at the national level**. Finally, it embraces a principle of economy and pragmatism: **do not reinvent the wheel.** There is no sense in constructing a wholly new framework from scratch when a recognized and well-tested one already exists. Accordingly, the Italian NCF takes the **NIST CSF** as its starting point, inheriting its structure of Functions, Categories, and Subcategories and adapting it to the Italian situation.

8.2 What the National Framework Is and Is Not

A precise understanding of the NCF's nature requires drawing a sharp boundary around what it is *not*. The National Framework is **not a standard**, and consequently **it is not certifiable**. This distinguishes it fundamentally from ISO 27001.

What it *is* instead is a **tool for self-assessment**. It allows an organization to assess its current cybersecurity posture—its **current profile**—and to define the posture it wishes to attain—its **target profile**. It then supports the organization

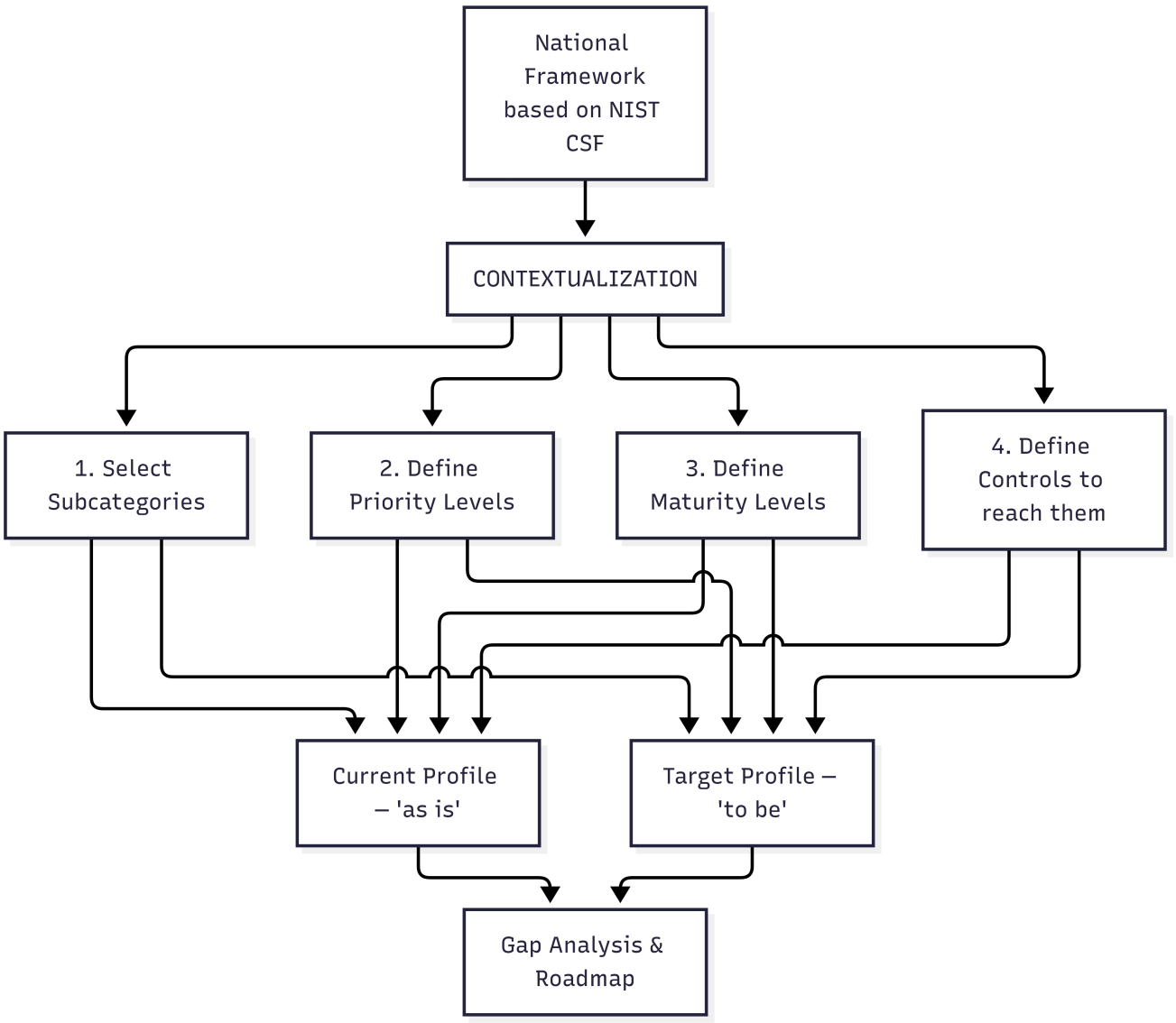
in performing the **gap analysis** between the two and in defining a **roadmap** to close that gap. In spirit and mechanism it is far closer to the NIST CSF than to ISO 27001: it organizes and communicates risk and guides improvement, rather than imposing auditable requirements leading to a certificate.

8.3 Contextualization

The central and distinctive mechanism of the Italian NCF is **Contextualization**: the process by which the framework is "customized" so as to adapt it to a specific context, whether a sector, a public administration, or an individual organization. Contextualization is the operation that turns a generic national framework into a concrete, tailored instrument.

A Contextualization is defined through four steps. First comes the **selection of Subcategories**—choosing, from the full catalogue inherited from the CSF structure, those that are relevant to the context at hand. Second comes the **definition of a Priority Level** for each selected Subcategory. Third comes the **definition of Maturity Levels**. Fourth comes the definition of the **Controls** needed to reach those maturity levels. In this way the framework layers two additional dimensions—priority and maturity—onto the Functions, Categories, Subcategories, and Informative References it inherits.

Once a Contextualization has been established, it becomes possible to define the two profiles familiar from the CSF: the **Current Profile**, representing the *"as is"* state of the organization, and the **Target Profile**, representing the *"to be"* state it aspires to. The gap between the two again drives the improvement roadmap.



The flexibility of Contextualization is deliberately broad. A Contextualization can be defined for a single organization; for an entire economic or productive sector; for groups of organizations grouped by scale, such as SMEs or large enterprises; for organizations belonging to regulated sectors, such as public administrations or banks; and for critical infrastructures. This range directly answers the structural concern noted earlier: because a Contextualization can be built specifically for SMEs, the framework can serve the overwhelming majority of the Italian economy without demanding the resources of a large enterprise.

[Photo: Contextualization architecture diagram] The architecture of the framework places the **National Cyber Security Framework** across the top as the common reference. On its left sits the **Implementation Tier** dimension, drawing on enterprise risk-management references such as CAS, COSO, and ISO. On its right sits the **Priority Levels / Maturity Levels** dimension, drawing on IT and security standards such as COBIT, SANS, and ISO. Between these two flanks, the central body of the diagram stratifies the economy into layers—Critical Infrastructures at the top, Large Organizations in the middle, and Small and

Medium Enterprises at the base—crossed vertically by production sectors and regulated sectors. The key theoretical message of the layout is that the single national framework can be contextualized simultaneously along two axes: *vertically* for a given productive or regulated sector, and *horizontally* as a general framework contextualization driven by enterprise risk management on one side and security standards on the other.

8.4 Contextualization for GDPR

A particularly significant Contextualization is the one that integrates **data protection** into cybersecurity. The reasoning is that cybersecurity and data protection, though historically distinct disciplines, converge on the same assets and the same risks. By combining them, the framework produces a unified **National Cybersecurity Framework and Data Protection**, and this combined instrument can then itself be contextualized for the **GDPR**—the European General Data Protection Regulation.



This integration is reflected in the evolution to **Framework 2.0**, whose architecture adds an explicit **Data Protection Regulations** dimension—with **GDPR** as its leading entry—alongside the existing IT and security standards. In other words, data-protection regulation is elevated to a first-class reference axis of the framework, standing on equal footing with the technical security standards that inform priority and maturity levels.

8.5 Framework 2.0: The Four Placement Cases

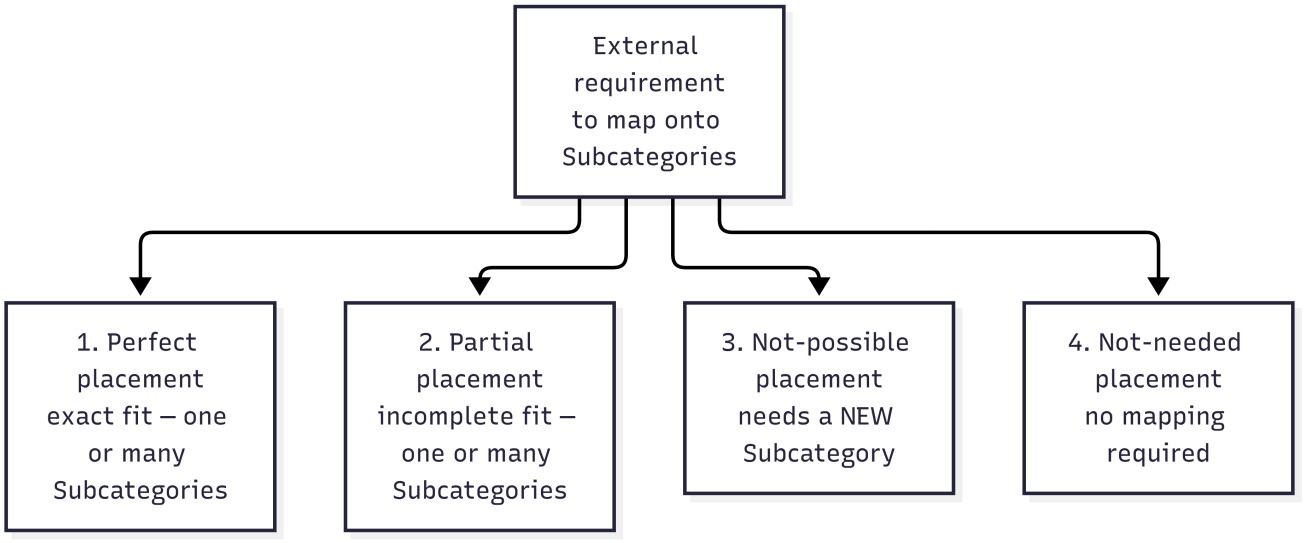
When an external requirement, control, or regulatory obligation must be mapped onto the framework's structure of Subcategories, four distinct cases can arise. Understanding these cases is essential to performing a Contextualization correctly, because they determine whether the existing structure suffices or must be extended.

The first case is **perfect placement**, in which the requirement maps cleanly and completely onto the existing structure. This may occur on a single Subcategory or across more than one Subcategory, but in either situation the fit is exact.

The second case is **partial placement**, in which the requirement maps onto the existing structure only incompletely—covering part but not all of a Subcategory. Again this may occur on a single Subcategory or on more than one.

The third case is **not-possible placement**, in which the requirement cannot be accommodated by any existing Subcategory at all. This situation signals a genuine gap in the framework's structure and creates the **need for a new Subcategory** to be added.

The fourth case is **not-needed placement**, in which a mapping is simply not required—the requirement does not call for placement within the framework.



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9. Implementation: ISO/IEC 27002

With the framework layer complete, the chapter now turns to the **implementation layer**—the instruments that answer the question of *how* to apply the strategy. The first of these is **ISO/IEC 27002**, the natural companion to ISO 27001.

9.1 Nature and Purpose

ISO/IEC 27002 is the **companion standard on information security controls**. Where ISO 27001 specifies *what* must be managed, ISO 27002 explains *how* the controls can be understood and implemented. It provides a reference set of

generic controls together with detailed implementation guidance, and it is designed to be used within an ISMS based on ISO 27001, for implementing commonly accepted controls, or for developing organization-specific guidance. In practical terms, it gives security teams the detailed direction they need to select and apply controls consistently.

A defining characteristic sets 27002 apart from 27001: **it is not a certification standard.** Certification is always against ISO 27001, never against ISO 27002. The reason is fundamental and worth dwelling on: **there is no single correct way to implement a control.** Because implementation legitimately varies according to an organization's circumstances, technologies, and risk profile, 27002 can offer guidelines and best practices but cannot impose the kind of uniform, verifiable requirements that certification demands. It describes the detail of each control, starting from Annex A of ISO 27001 and expanding each entry into actionable guidance.

9.2 Structure of ISO/IEC 27002:2022

The 2022 edition reorganized the control catalogue into **four themes**, or macro-areas, a simplification intended to make the catalogue easier to navigate and apply. The total control set comprises **93 controls**, distributed across the four themes as follows.

The **Organizational** theme contains **37 controls**, covering governance, policies, supplier relationships, incident management, and related management practices. The **People** theme contains **8 controls**, addressing matters such as screening, awareness, training, and the responsibilities of personnel. The **Physical** theme contains **14 controls**, concerning the protection of facilities, equipment, and physical environments. The **Technological** theme contains **34 controls**, covering access control, logging, malware protection, secure development, network security, and related areas.

Theme	Number of controls
Organizational	37
People	8
Physical	14
Technological	34
Total	**93**

An instructive observation emerges from comparing these themes. The **People** controls tend to be nearly the same for every organization—background screening, training, and disciplinary processes look broadly similar regardless of industry—whereas the **Technological** controls may vary considerably from one organization to another, reflecting differences in systems, architectures, and technical environments.

The individual controls within each theme can be surveyed to convey the catalogue's breadth. The **Organizational** controls (5.1–5.37) include policies for information security; information security roles and responsibilities; segregation of duties; management responsibilities; contact with authorities and with special interest groups; threat intelligence (new in 2022); information security in project management; inventory of information and assets; acceptable use and return of assets; information classification and labelling; access control and identity management; supplier security policies; information security in cloud services (new); incident management and ICT continuity; and legal, compliance, documentation, and audit requirements.

The **People** controls (6.1–6.8) comprise screening (background verification of candidates prior to employment); terms and conditions of employment; information security awareness, education, and training; a disciplinary process for handling policy violations; responsibilities that remain after termination of employment; confidentiality or non-disclosure agreements; remote-working security controls; and a mechanism for the prompt reporting of security events.

The **Physical** controls (7.1–7.14) cover physical security perimeters; physical entry controls; securing offices, rooms, and facilities; physical security monitoring (new); protection against physical and environmental threats; working in secure areas; clear desk and clear screen; equipment siting and protection; security of assets off-premises; storage media; supporting utilities; cabling security; equipment maintenance; and secure disposal or re-use of equipment.

The **Technological** controls (8.1–8.34) are conveniently grouped into four clusters. *Access and Identity* (8.1–8.6) includes user endpoint devices, privileged access rights, access control, authentication, identity management, and privileged utilities. *Data Protection* (8.10–8.12, 8.24) includes information deletion (new), data masking (new), data leakage prevention (new), and cryptography. *Security Operations* (8.15–8.22) includes logging and monitoring, clock synchronization, web filtering (new), network security, configuration management (new), and vulnerability management. *Secure Development and Acquisition* (8.25–8.34)

includes the secure development lifecycle, secure coding (new), security testing, outsourced development, and separation of environments.

9.3 Annex A versus ISO 27002: Normative versus Guidance

The precise relationship between ISO 27001's Annex A and ISO 27002 is the key to understanding how the two standards interlock, and it rests on a single distinction: **Annex A is normative; ISO 27002 is guidance.**

Annex A of ISO 27001 contains a catalogue of control objectives and controls that organizations select as part of their ISMS. It is **normative**—binding and linked directly to the risk-treatment process—but it is also terse, typically stating only the control's name and a one-sentence statement of what the control requires. ISO 27002:2022, by contrast, provides **detailed implementation** guidance, purpose statements, and attributes for the *same 93 controls* referenced in ISO 27001, helping translate the austere Annex A entries into actionable operative measures.

The difference is vividly illustrated by a single control. In Annex A of ISO 27001, the control on access to program source code appears as little more than a title and the normative statement that access to program source code shall be restricted. In ISO 27002, the same control is accompanied by extensive implementation guidance: that source code and associated items such as designs, specifications, and verification plans should be strictly controlled to prevent the introduction of unauthorized functionality and to preserve confidentiality; that source libraries should preferably not be held in operational systems; that source code and libraries should be managed according to established procedures; that support personnel should not have unrestricted access; that updates and the issuing of source to programmers should occur only after appropriate authorization; that program listings should be held securely; that an audit log of all accesses should be maintained; that maintenance and copying should be subject to strict change-control procedures; and that, where source code is to be published, additional integrity controls such as digital signatures should be considered. The same control thus exists in both documents, but Annex A declares the *obligation* while ISO 27002 supplies the *method*.

10. Implementation: NIST SP 800-53

The second instrument of the implementation layer is **NIST Special Publication 800-53**. If ISO 27002 is the implementation companion to the ISO

27001 management-system standard, SP 800-53 plays an analogous role on the NIST side, translating high-level framework outcomes into a concrete, detailed catalogue of controls—though it does so with far greater granularity and a distinctly American, federal orientation.

10.1 The Special Publications Series

NIST periodically publishes technical reports that provide guidelines, recommendations, and reference material, particularly in the field of information security. These are known as **Special Publications (SPs)**. The **SP 800 series** is NIST's primary channel for publishing computer, cyber, and information security guidelines, recommendations, and technical specifications. SP 800-53 is one member of this series, and it is among the most important.

NIST Special Publication 800-53 provides a comprehensive catalogue of **security and privacy controls** for information systems and organizations. Its controls are deliberately **flexible and customizable**: organizations select and tailor them according to their mission, business needs, risk tolerance, and operating environment. Crucially, like ISO 27002, SP 800-53 is **a control catalogue and implementation reference, not a certification standard** such as ISO 27001. It supplies the raw material of controls; it does not issue certificates.

The publication pursues three explicit aims. The first is to provide a **catalogue of security and privacy controls** for federal organizations and information systems. The second is to provide a **process to select controls** that protect an organization's mission, functions, image, reputation, assets, and individuals—and the state itself—from cyberattacks, disasters, malfunctions, and human errors. The third is to provide a **methodology for selecting critical controls**, which is the concept of the **baseline**.

10.2 Key Concepts

Several concepts underpin the structure of SP 800-53 and recur throughout its use. **Security controls** are the safeguards and countermeasures that protect the confidentiality, integrity, and availability of systems and information. **Privacy controls** are the administrative, technical, and physical safeguards that manage privacy risks and support compliance with applicable privacy requirements. **Control families** are groups of controls organized by topic—for instance, access control, incident response, risk assessment, and supply-chain risk management. **Control enhancements** are additional requirements that strengthen or extend a

base control for higher-risk environments. **Tailoring** is the process of adapting controls to a specific organization, system, mission, technology, or risk profile.

10.3 Revision 5 and the Twenty Control Families

The current edition, **Revision 5**, marks a significant structural change: it **integrates security and privacy controls into a single consolidated catalogue**, rather than treating them as separate concerns. This consolidation reflects the growing recognition that security and privacy are deeply intertwined.

Revision 5 organizes the catalogue into **20 control families**. These families span the full breadth of organizational security and include Access Control, Audit and Accountability, Identification and Authentication, Incident Response, Risk Assessment, System and Communications Protection, and Supply Chain Risk Management, among others. Each family contains the security controls related to its general security topic, and the organization into families exists specifically to ease the control-selection and specification process.

The twenty families, each denoted by a two-letter identifier, are the following:

ID	Family	ID	Family
AC	Access Control	PE	Physical and Environmental Protection
AT	Awareness and Training	PL	Planning
AU	Audit and Accountability	PM	Program Management
CA	Assessment, Authorization, and Monitoring	PS	Personnel Security
CM	Configuration Management	PT	PII Processing and Transparency
CP	Contingency Planning	RA	Risk Assessment
IA	Identification and Authentication	SA	System and Services Acquisition
IR	Incident Response	SC	System and Communications Protection
MA	Maintenance	SI	System and Information Integrity
MP	Media Protection	SR	Supply Chain Risk Management

Within this structure, each control generally includes a **base control**, a **discussion**, **related controls**, **control enhancements**, and **references**. As with the NIST CSF Core, the controls are intended to be **outcome-oriented and adaptable** rather than implemented as an identical checklist in every organization —a philosophy consistent across all the NIST instruments.

10.4 The Structure of a Security Control

Each security control in SP 800-53 follows a consistent internal structure composed of five components, and understanding this anatomy is essential to reading and applying the catalogue.

The **Control** section prescribes the specific security-related activities or actions to be carried out by organizations or by information systems. The **Supplemental Guidance** section provides non-prescriptive, additional information that helps interpret and apply the control. The **Control Enhancements** section provides statements of additional security capability that either add functionality and specificity to the base control or increase its strength, for use in higher-risk environments. The **References** section lists the applicable federal laws, Executive Orders, directives, policies, regulations, standards, and guidelines relevant to the control. Finally, the **Priority and Baseline Allocation** section provides both the recommended priority codes used for sequencing implementation decisions and the initial allocation of controls and enhancements to the baselines.

This structure can be made concrete with a representative control drawn from the Audit and Accountability family, **AU-3, Content of Audit Records**. Its *Control* statement requires that the information system generate audit records establishing what type of event occurred, when and where it occurred, the source and outcome of the event, and the identity of any individuals associated with it. Its *Supplemental Guidance* clarifies that the content satisfying this requirement may include time stamps, source and destination addresses, user or process identifiers, event descriptions, success or failure indications, filenames involved, and the access-control or flow-control rules invoked. Its *Control Enhancements* add, among others, the capability to generate records containing additional organization-defined detailed information, and the capability to provide centralized management and configuration of the content captured in audit records. Its *References* in this instance are none. Its *Priority and Baseline Allocation* assigns the control priority P1 and allocates the base control AU-3 to the LOW baseline, AU-3 with enhancement (1) to the MODERATE baseline, and AU-3 with enhancements (1) and (2) to the HIGH baseline. This single example demonstrates

how the five components work together to move from a terse requirement to a fully specified, risk-calibrated control.

10.5 Baselines: SP 800-53B

The mechanism for selecting controls in a principled way is provided by a **companion document, SP 800-53B**, devoted to the concept of the **security control baseline**. Its aim is to assist organizations in making the appropriate selection of controls for their information systems.

Baseline controls are the starting point for the control-selection process. They are chosen on the basis of the **security category** and associated **impact level** of the information system in question. Three baselines have been identified, corresponding to **low-impact, moderate-impact, and high-impact** information systems. An organization first categorizes its system by impact level and then adopts the matching baseline as its point of departure, afterward tailoring the controls to its specific circumstances.

The baseline concept carries notable **limitations** that bound its general applicability. It is tightly tied to the notion of *federal* information systems and to U.S. legislation, policies, directives, instructions, regulations, and memoranda—more than fifty such instruments in total, all U.S.-based. It is also **very complex** and oriented toward operational personnel rather than strategic decision-makers. These limitations explain why SP 800-53, for all its thoroughness, is best understood as a deeply detailed federal control catalogue rather than a lightweight, universally portable instrument.

A worked scenario clarifies how baselines operate in practice. Suppose an organization runs an internal web application for a federal agency that processes sensitive but unclassified data, where a breach would cause **serious—though not catastrophic—impact** on confidentiality, integrity, or availability. Under FIPS 199, such a system is categorized as **Moderate-impact**. The organization would then turn to SP 800-53B and select the **Moderate security control baseline**, which includes roughly **287 security controls and enhancements** drawn from SP 800-53. From this starting point it would tailor the set to its particular environment.

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11. Implementation: CIS Controls

The third and final instrument of the implementation layer is the **CIS Controls**, published by the **Center for Internet Security (CIS)**, a U.S. non-profit organization focused on improving global cybersecurity standards and practices. CIS collaborates with experts drawn from industry, government, and academia to develop best practices and guidelines that help organizations secure their IT systems and data against cyber threats. Where SP 800-53 is exhaustive and federal, the CIS Controls are prioritized and pragmatic, designed to tell an organization not merely *what* could be done but *what to do first*.

The CIS Controls are a **prioritized set of actions** designed to help organizations strengthen their cybersecurity posture. Developed by a global community of cybersecurity experts, they provide clear, actionable steps to protect against the most prevalent threats. They are organized into categories and **ranked by effectiveness and ease of implementation**, which makes them exceptionally practical for organizations of every size.

11.1 Design Principles and Structure

Three design principles govern the construction of the CIS Controls. The first is that **offense informs defense**: controls are selected and prioritized using *empirical knowledge* of how attackers actually behave and how their behavior can be stopped. The second is **focus and feasibility**: each safeguard is specific and practical to implement, deliberately avoiding "nice-to-have" or purely theoretical requirements. The third is that the controls are **measurable and aligned**: safeguards are written to be measurable and to align with other frameworks, including the NIST CSF, ISO 27001, OWASP, and MITRE ATT&CK.

Structurally, **CIS Controls v8.1** consists of **18 top-level controls** broken down into **153 safeguards**, each describing a single actionable requirement. The eighteen controls are grouped into three tiers of ascending sophistication. The **Basic Controls (1–6)** cover fundamental cybersecurity measures. The **Foundational Controls (7–16)** cover more advanced security practices. The **Organizational Controls (17–18)** cover the policies and processes that govern security. A defining feature of the scheme is that the **controls are numbered by priority**: the numbering is not arbitrary but reflects the order in which controls should be addressed.

[Photo: The 18 CIS Controls] The eighteen controls, arranged around a central lock emblem, are: (1) Inventory and Control of Enterprise Assets; (2) Inventory and Control of Software Assets; (3) Data Protection; (4) Secure Configuration of

Enterprise Assets and Software; (5) Account Management; (6) Access Control Management; (7) Continuous Vulnerability Management; (8) Audit Log Management; (9) Email and Web Browser Protections; (10) Malware Defenses; (11) Data Recovery; (12) Network Infrastructure Management; (13) Network Monitoring and Defense; (14) Security Awareness and Skills Training; (15) Service Provider Management; (16) Application Software Security; (17) Incident Response Management; and (18) Penetration Testing. The theoretical point conveyed by the arrangement is that the lowest-numbered controls address the most fundamental and highest-priority hygiene—knowing what assets you have—while the higher numbers progress toward advanced and organizational practices.

11.2 Implementation Groups

To make the 153 safeguards usable by organizations of radically different capacity, CIS assigns them to **Implementation Groups (IG)**—IG1, IG2, and IG3—based on an organization's size, risk, and resources.

IG1, described as **basic cyber hygiene**, comprises approximately **56 safeguards** that *every* organization should implement, focused on the most common attack vectors. **IG2 and IG3** progressively add more advanced safeguards for organizations with higher risk, more complex environments, or regulatory obligations.

The IG1 safeguards cover the essentials of defensive hygiene: maintaining an **inventory** of devices and software (Controls 1–2); applying **secure configuration** of systems and applications (Control 4); enforcing **multi-factor authentication** and least privilege for accounts and access (Controls 5–6); performing **patch management and vulnerability management** (Control 7); and sustaining **backup and tested data recovery, basic logging, anti-malware defenses, and awareness training** (Controls 8–14). These are the **first controls to implement** when building or improving a security program, a principle CIS captures in its slogan: **"start with IG1."**



11.3 Mapping to Other Frameworks and the CIS Workflow

A major practical virtue of the CIS Controls is their **interoperability**. CIS Controls v8.1 are mapped to and referenced by multiple legal, regulatory, and policy frameworks, including NIST CSF 2.0, ISO 27001, PCI DSS, SOC 2, and CMMC. The addition of a Govern function and updated mappings in v8.1 improves consistency specifically with the NIST CSF 2.0 functions. The consequence is powerful: an organization can **implement the CIS Controls once and then use the mappings to demonstrate alignment across several governance and compliance regimes** simultaneously, avoiding duplicated effort.

This leads to a recommended **CIS Controls workflow**. First, use the CIS Controls as a prioritized roadmap, starting with the IG1 safeguards and extending to IG2 and IG3 according to the organization's risk profile and regulatory drivers. Second, treat each safeguard as a measurable requirement backed by evidence of both its *design* and its *operation*, rather than as a generic "good practice." Third, use the Controls to structure the organization's security governance, and then plug in the other frameworks—NIST CSF, ISO 27001, the Italian NCF—as overlays and mappings for reporting purposes.

11.4 Benefits of the CIS Controls

The advantages of the CIS Controls follow directly from their design. They offer **prioritized, actionable steps**, focusing effort on the most impactful risk-reduction measures. They deliver genuine **attack-surface and risk reduction** by directly targeting common attack vectors such as unmanaged assets, misconfigurations, missing patches, and weak access control. Through IG1 they establish an **essential cyber-hygiene baseline** that provides a realistic minimum program for an organization of any size. They enable the **efficient use of limited resources**, helping allocate scarce budget and staff to the highest-value controls first. They provide **multi-framework alignment**, mapping cleanly to NIST CSF, SP 800-53, ISO 27001, PCI DSS, and HIPAA, which eases compliance. And they constitute a **widely recognized "reasonable" standard**, used globally and increasingly cited by regulators and insurers as evidence of due diligence.

11.5 Treating the WannaCry Risk

The CIS Controls bring the WannaCry narrative to its operational conclusion. Having, through the earlier frameworks, *identified* the central risk—that unpatched Windows systems could be exploited and cause widespread compromise—and having *decided* that the risk must be treated, the security team faces the final, concrete question: which controls to actually implement, given a single month in which to act.

The appropriate measures are precisely those drawn from the prioritized CIS hygiene: a **formal patch-management process**, a complete **asset inventory**, **network segmentation**, and **vulnerability scanning**. These directly dismantle the conditions that made the attack possible. By contrast, measures such as a new company logo or an annual security presentation, whatever their other merits, do nothing to reduce the technical risk at hand and would be mis-prioritized responses to the threat.

A final and subtle lesson concerns the difference between *claiming* a control and *evidencing* it. Suppose the organization asserts that it performs vulnerability scanning regularly. What would actually convince an auditor that this control is implemented? Statements such as "we have a vulnerability-management policy," "the CISO says scanning is performed," "a scanner is installed," or "the vendor says the scanner works" are all insufficient, because a policy is not its execution, an assertion is not proof, and an installed or vendor-endorsed tool is not evidence of operation. What genuinely demonstrates the control are **scan reports from the last six months** together with **evidence that identified vulnerabilities were assigned and remediated**. This distinction—between the *design* of a control and its demonstrable *operation*—is the heart of auditability, and it is exactly the discipline whose absence allowed a known, patchable vulnerability to bring hospitals to a standstill.